



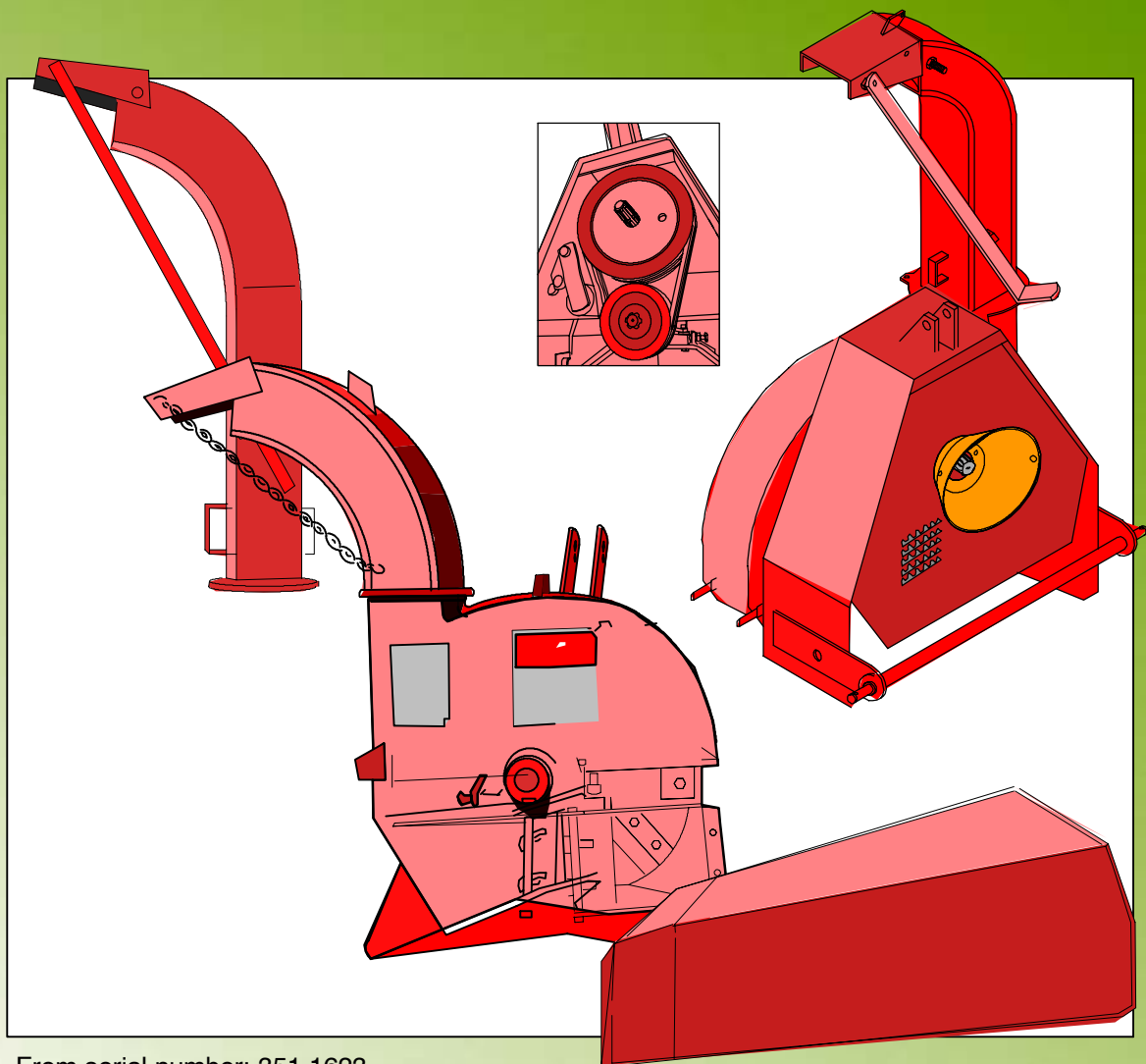
CH 160

OPERATION, MAINTENANCE AND SPARE PARTS

FARMII

CHIPPER CH 160

– CH 160T $i=2,1$



From serial number: 351 1623

READ THIS OPERATION AND MAINTENANCE MANUAL CAREFULLY
BEFORE USING THE CHIPPER

Farmi Forest Corporation

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CH 160

WARNING SYMBOLS IN THIS MANUAL



Imminent danger which could cause serious personal injury or death.



Danger which could cause personal injury.



Conditions or misuse that could damage equipment or machinery.

NOTE!

Reminders, such as for performing checks or carrying out maintenance or repair procedures.

INTRODUCTION

This manual includes the information and maintenance instructions required for operating the machine in the optimal manner.

Although you have experience in using this kind of machinery, read the operation and maintenance instructions carefully since they include information enabling efficient and safe operation. Regular maintenance is the best way to guarantee the efficient and economical performance of the machine.



Each and every operator must read, understand, and follow all safety instructions and procedures.

CUSTOMER FEEDBACK

We are happy to receive your opinions and suggestions for improvements by mail, fax or e-mail. All implemented suggestions for improvements will be rewarded.

Farmi Forest Oy

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MANUFACTURER'S DECLARATION OF CONFORMITY

We, Farmi Forest Corporation,
Ahmolantie 6, FIN-74510 PELTOSALMI, Finland,

hereby declare that the machine

Farmi

(make)

CH160 / CH160T

(type)

(serial number)

conforms to the directives 98/37/EC and 89/366/EC, as amended, and the national regulations bringing these directives into force.

In designing the machine, the following unified standards have been applied:

EN292-2, EN294, EN 982, EN60204-1

In designing the machine, the following national standards and specifications have been applied:

prEN 13525, prEN1553-1

Peltosalmi,

(place)



Juha Hallivuori

01.09.2002

(date)

Farmi Forest Oy

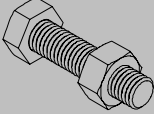
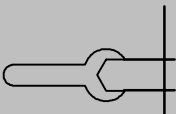
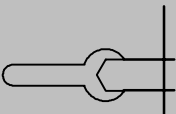
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SPARE PARTS 24

 Thread	 Millimeter	 Inch
M6	10	7/16
M8	13	1/2
M10	17	11/16
M12	19	3/4
M16	24	15/16
M20	30	1 3/16

Tilatessasi varaosia, ilmoita koneen tyyppi konekilvestä, varaosan tilausnumero, osan nimi, sekä tilattava kappalemäärä.

Esimerkki: CH 160, 43510220, Terä, 2 kpl

1. SAFETY

1.1. GENERAL

These safety instructions are intended for personnel operating, servicing, and repairing FARMI equipment.

These instructions will help you to:

- operate the machine safely
- operate the machine appropriately and effectively
- notice, avoid, and prevent potentially hazardous situations

The manufacturer has supplied safety instructions for the machine. The instructions must be kept in the compartment of the machine that is reserved for the instructions, and they must always be available on the site where the machine is used. Before using the machine, each and every operator of the machine must read the safety, maintenance, and operation instructions. These instructions must always be followed.

The machine meets the specified technical standards as well as required and appropriate safety regulations.

In addition to those in the manual, the instructions of the local occupational safety authorities as well as national laws and regulations must be followed.

Using the machine for tasks other than those for which it is intended or exceeding its performance is not appropriate use of the machine. The manufacturer/supplier is not responsible for damage resulting from this kind of usage.

1.2. OPERATIONAL SAFETY



Occupational accidents often take place in abnormal or nonstandard situations. Consider all possible, even unlikely, situations that could occur during each task.

- Due to design considerations, the machine's noise level may exceed 85 dB; wear hearing protection and other appropriate safety equipment.
- Ensure that occupational safety regulations are followed when working.
- Familiarize yourself with the machine's risk zones and prevent unauthorized persons from entering these zones.
- During operation, the operator of the machine is responsible for the safety and risk zones for the entire work site.
- Each time the machine is to be used, first check its operational safety and correct any deficiencies.
- Before starting and using the machine, ensure that it does not pose a danger to other persons or a risk of damage.
- Exceeding the specified performance ratings is prohibited.
- Do not operate the machine while sick or under the influence of alcohol or drugs.
- All of the machine's safety and warning labels must be legible and undamaged. Replace if necessary.
- The operator of the machine must always have unobstructed visibility on-site. If necessary, an assistant must be provided to the operator.
- In the event of any defect compromising operational safety, stop the machine!
- If it is not possible to correct the impairment to operational safety, work must be stopped.

1.3. OPERATION

- Familiarize yourself with the capacity and limitations of the machine and equipment, as well as how they can be used most efficiently.
- Familiarize yourself carefully with the operation and functions of the machine before operating it.
- The operator of the machine must also be familiar with the operation of the accessories.
- Do not stand unnecessarily close to the machine during operation.
- Prevent unauthorized persons from entering the working area during operation.
- The operator of the machine must always have unobstructed visibility of the site.
- Operating a damaged or defective machine is prohibited.
- Ensure the following:
 - That the safety and protective devices of the machine are not removed or modified and that they are used properly.
 - Regular safety tests and similar inspections are carried out.
- Remember that the manufacturer of the machine is not responsible for damage that is caused by:
 - incorrect, careless, or inappropriate use
 - misuse caused by errors on the part of untrained personnel
- When leaving the machine, always secure it from unauthorized and unintended operation.
- When parking, ensure that the machine is properly supported.
- Avoid placing sudden, demanding loads on the machine.
- Use the controls carefully, especially in cold conditions.

1.4.OILS AND GREASES

- Always use the oil grades recommended by the manufacturer.
- Use the recommended oils in cold conditions.
- Never mix different fluids or oils with each other.
- Follow the manufacturer's lubrication instructions.

1.5.MAINTENANCE

- Modifications without the manufacturer's permission are prohibited.
- Remember that the manufacturer of the machine is not responsible for damage that is caused by:
 - modifications made without the manufacturer's permission
 - use of non-original spare parts
 - normal wear
- Safety devices must be re-fitted immediately after maintenance.
- Never try to take hold of moving parts when the engine is running.
- Ensure that the machine cannot be started during maintenance.
- Electrical and hydraulic failures are to be repaired by authorized personnel only. The same applies to the welding of load-bearing structures.
- Follow the maintenance intervals listed, and perform the indicated annual safety inspections.

1.6.LIFTING

- Lift only with the proper type of lifting device, and ensure that it has an appropriate lifting capacity.
- Ensure that the lifting equipment is in good working condition. The lifting equipment must be inspected regularly.
- Ensure that you know the weight of the load to be lifted and never exceed the lifting capacity stated by the manufacturer of the lifting device.
- Select routes for lifting that pose no danger.
- Lifting points are marked on the machines with stickers as shown in the figure. Use only the marked points for lifting the machine.

1.7.STORAGE

- If the chipper is to be stored for a long period, lubricate the knives with, for example, petroleum jelly.
- Ensure that the water drainage holes in the lower chamber are unobstructed.

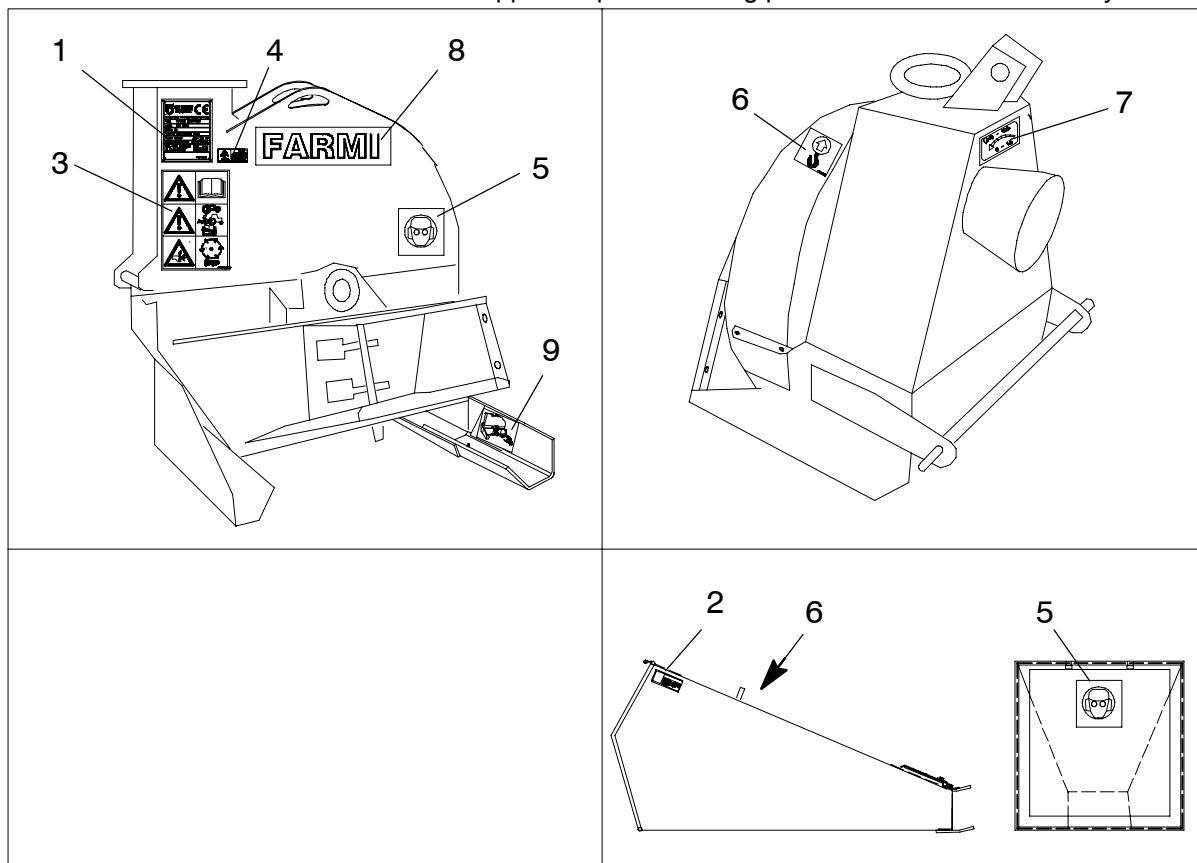
2.SAFETY INSTRUCTIONS FOR HYDRAULIC SYSTEMS

1. Hydraulic systems must be serviced by qualified, experienced hydraulics mechanics only.
2. Check the machine for any leaks and repair them before using the machine.
3. Clean the area to be repaired. Do not use solvents to clean components.
4. Never repair the hydraulic system when it is pressurized.
5. Support the machine before servicing or making repairs. Do not detach the cylinders or their valves before supporting the machine.
6. When replacing hydraulic components and hoses, use original spare parts or parts recommended by the manufacturer. In particular, ensure that the pressure classes correspond to the operating pressure.
7. If oil gets in the eyes, rinse them thoroughly with water and consult a doctor.
8. Oil can be removed from skin by wiping and washing. Use safety equipment when handling oils and greases.
9. Prevent hydraulic oil from escaping into the environment. Waste oil must always be collected in a container to keep the environment clean!
10. When working in ecologically sensitive areas, use bio-oils.
11. Keep oil indoors in closed, original containers. Pour the oil into the tank directly from the container.
12. All caps, funnels, screens, and filling holes must be clean.

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3. STICKERS AND PLATES

These plates and stickers must be found on the chipper. Replace missing plates or stickers immediately.



T1

Farmi Forest Corporation Ahmolan tie 6 FIN-74510 PELTOSALMI FINLAND		
TYPE	FARMi CHIPPER	
MODEL	CH160 T	
SERIAL NO.		
YEAR OF MANUFACTURE	20	
POWER NEEDED	18–40 kW	
WEIGHT (HYDR/MANUAL)	450/330 kg	
MAX. HYDR. PRESSURE	175 bar	
OIL FLOW NEEDED	20 l/min	
		41011740

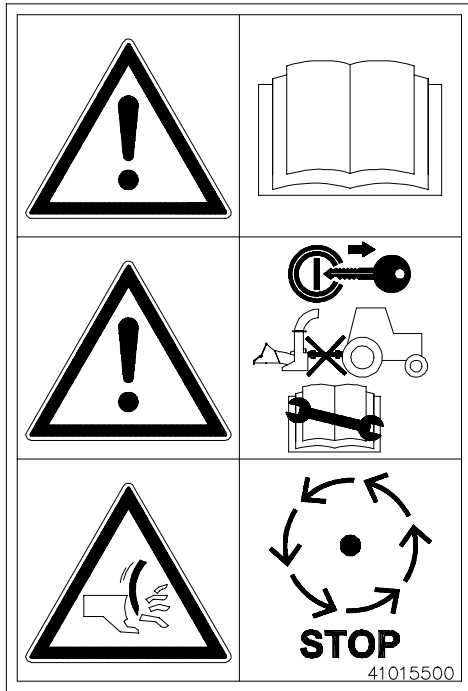
Number 1. Machine plate (41011740).

	valmistaja manufacturer
	Farmi Forest Oy FIN-74510 PELTOSALMI FINLAND

Number 2. Manufacturer sticker (40605214).

T2

CH 160



Number 3.Note! (41015500)

Note!
See operation and maintenance manual..

Note!
Stop the engine and remove the PTO shaft for maintenance.

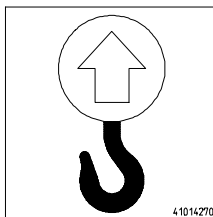
Risk of cutting injury!
Stop the engine and wait until the disk has stopped.



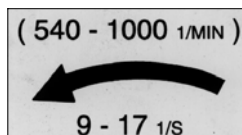
Number 4.GS sticker (51971919).



Number 5.Hearing protection symbol.
Wear hearing protection. (40142080). 2pcs.



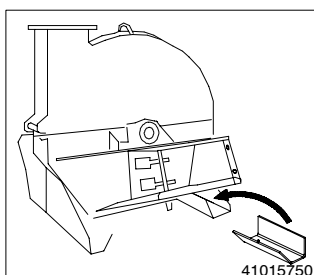
Number 6.Lifting point sticker (41014270). 2 pcs.



Number 7.Rotation speed sticker (40141160).



Number 8.“FARM” –sticker (20140489).



Number 9.Foot mounting instructions –sticker (41015750).

CH 160

3. PRESENTATION

The Farmi CH160 is a double – knife disk cutter for chipping wood up to Ø160 mm, for processing wood waste from, e.g., the side of jogging tracks or roads, and for other environmental chipping duties. The chipper can be driven by a tractor with a rating of 10–40 kW. The chipper can also be driven by a separate hydraulic motor.

CH160 accessories:

- Single – motor hydraulic feeder, mechanical or gravity hopper;
- Independent hydraulic unit HD11;
- Separate hydraulic motor drive;
- “Wille” adapter;
- “Bobcat” adapter;
- Belt transmission 160 T

4. MAIN COMPONENTS

1. *UPPER CHAMBER*
2. *LOWER CHAMBER*
3. *DISK*
4. *KNIFE*
5. *VERTICAL ANVIL*
6. *HORIZONTAL ANVIL*
7. *TWIG BLADE*
8. *DISCHARGE PIPE*
9. *LID*
10. *HYDRAULIC MOTOR (OPTIONAL)*
11. *BELT TRANSMISSION 160 T (OPTIONAL)*

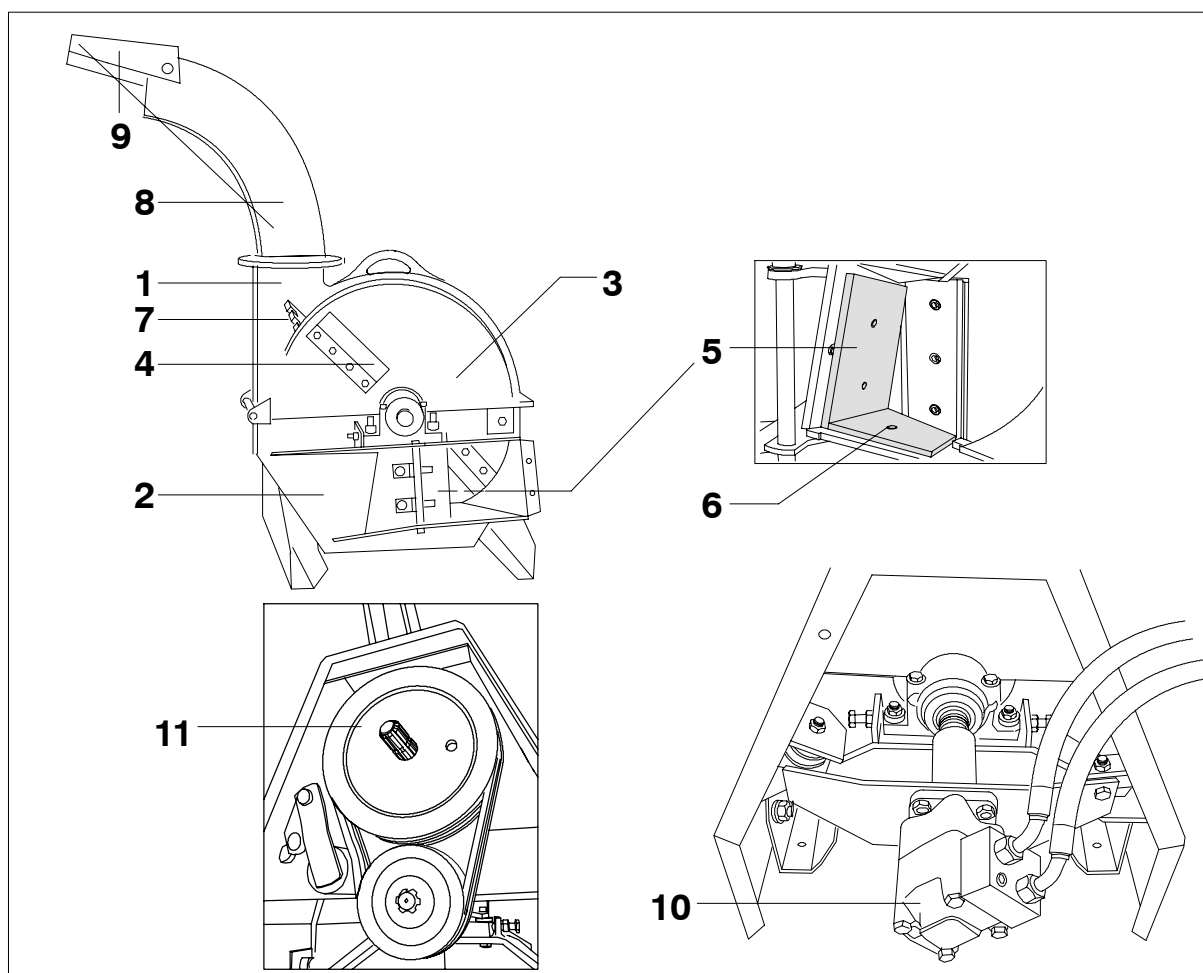


Fig. 1. Main components.

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5.DIMENSIONS

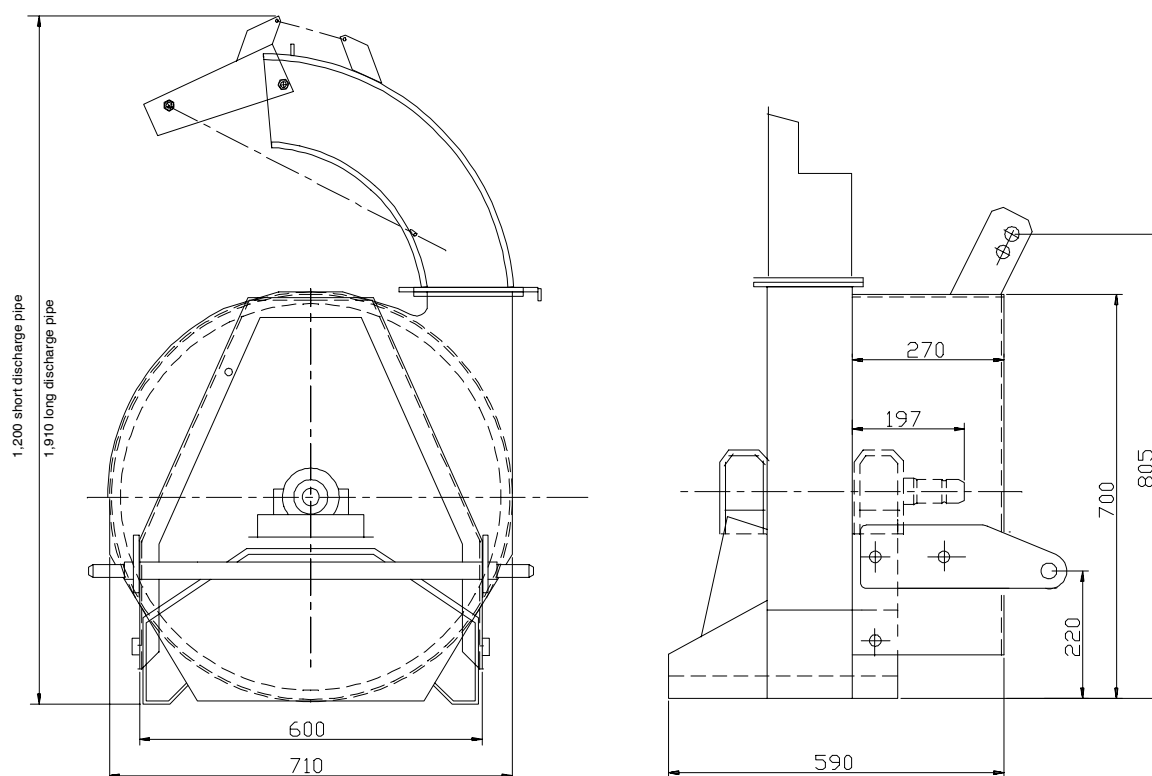


Fig. 2.CH160 dimensional drawing

6. TECHNICAL DATA

CH 160	
Type	disc chipper
Output	5-20 m ³ /h
Chip length	13 mm
Max. wood diameter	160 mm
Power demand	10 -40 kW
PTO RPM	540 or 1000 rpm
Number of knives	2
Power source	tractor or a separate hydraulic motor
Mounting	3-point
Chipper weight	CH 160 240 kg CH 160T 284 kg
Disc diameter	670 mm
Disk weight	100 kg
Discharge pipe	360° turning
Opening of upper chamber	to one side
Feeder	manual chute or hydraulic feeder
Sound pressure level	102 dB (A)
Sound power level	120 dB (A)
CEN/TC144 WG8N16	

Table 1. Technical data.

CH 160

5. LIFTING



Lifting points for each machine are marked with hook symbols.

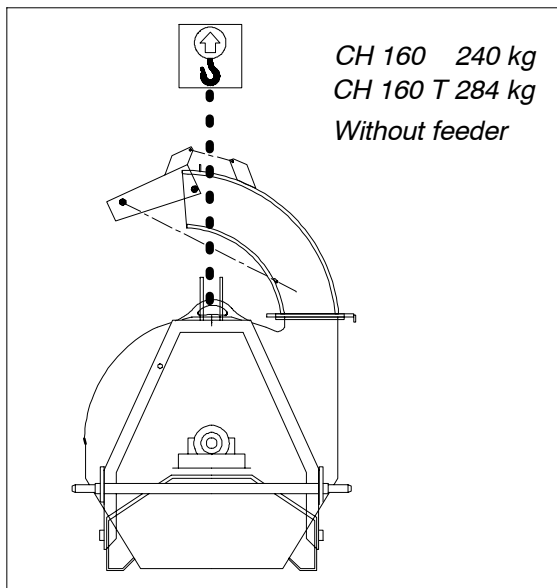


Fig. 3. CH160 Lifting point.

6. ASSEMBLY INSTRUCTIONS

1. Attach the feeder to the chipper from the left side with the hinge pin (1) and from the right side with two bolts (M12) (2). Remember to install the securing bolts (M6) (3). See Fig. 4.

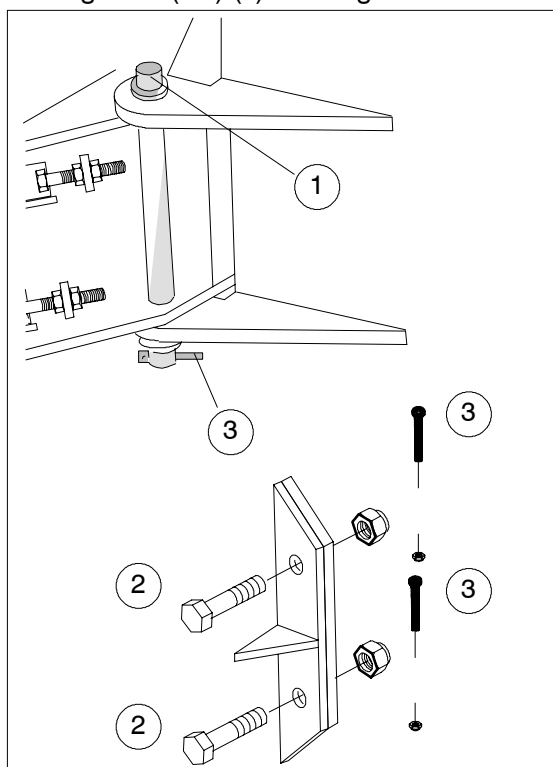


Fig. 4. Mounting the feeder.

2. Attach the discharge pipe to the chipper with two M10 bolts and one M16 bolt.
3. Mount the chipper on the tractor's 3-point hitch.
4. Install the PTO shaft.

5. Check that the length of the PTO shaft is correct for different hitch positions. For instructions on cutting the shaft, see Section 6.1.

6.1. CUTTING THE PTO SHAFT



The use of too long a PTO shaft with a 3-point hitch could damage the chipper's bearings or the tractor's PTO shaft. Both halves of the PTO shaft must be shortened equally.

Recommended PTO shafts:

Bondioli & Pavesi A6 + (RA1 with overrunning clutch.)

Walterscheid W2400 + (F3 with overrunning clutch.) If the tractor features a PTO brake, an overrunning clutch must be used (e.g., Ford).

1. Mount the machine on the tractor's 3-point hitch.
2. Lift the machine with the hitch so that the distance between the splined shafts is at its shortest.

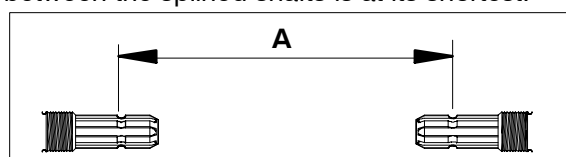


Fig. 5. Measure distance A when the distance between the splined shafts is at its shortest.

3. First, cut the thicker shield to the correct length (1). Remember the 20 mm clearance. Then, cut a similar length from the profile tube (2). Shorten the other PTO shaft half similarly. Remove burrs using a file.

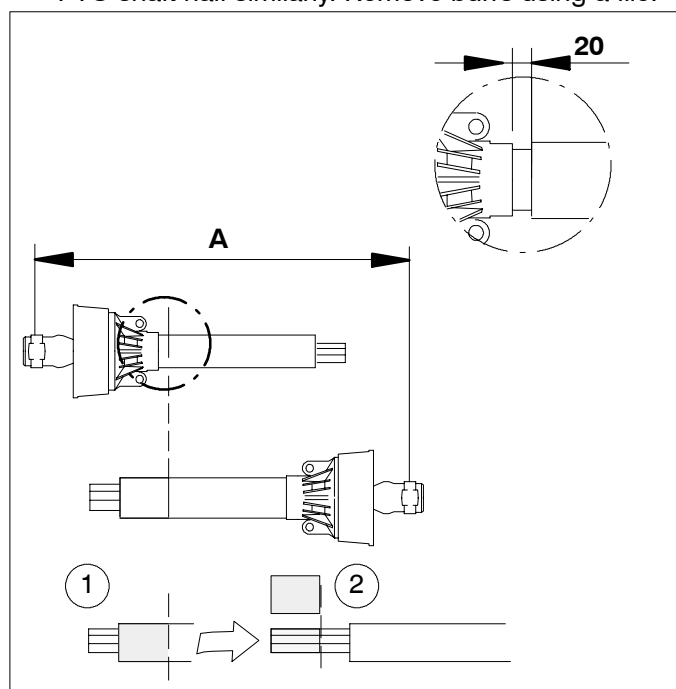


Fig. 6. Shortening the PTO shaft.

4. Insert the inner tube in the outer tube and carefully move the machine up and down. Ensure that the shaft was shortened sufficiently. Check that the shaft has a 20 mm clearance. In addition, move the machine sideways and ensure that the PTO shaft does not bottom.

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6.2.MOUNTING ON THE WILLE 445 ADAPTER (Mounting kit 43510500)

- Attach the top link brackets of the adapter and chipper together with M20 pin and nut. (1)
- Attach the grabbers to the chipper's drawbar assy with four bolts (M16) (2).
- The grabbers have three grooves for the drawbar of the drawbar assy. Attach the grooves of the grabbers (3) so that the chipper is as vertical as possible at the operating position.

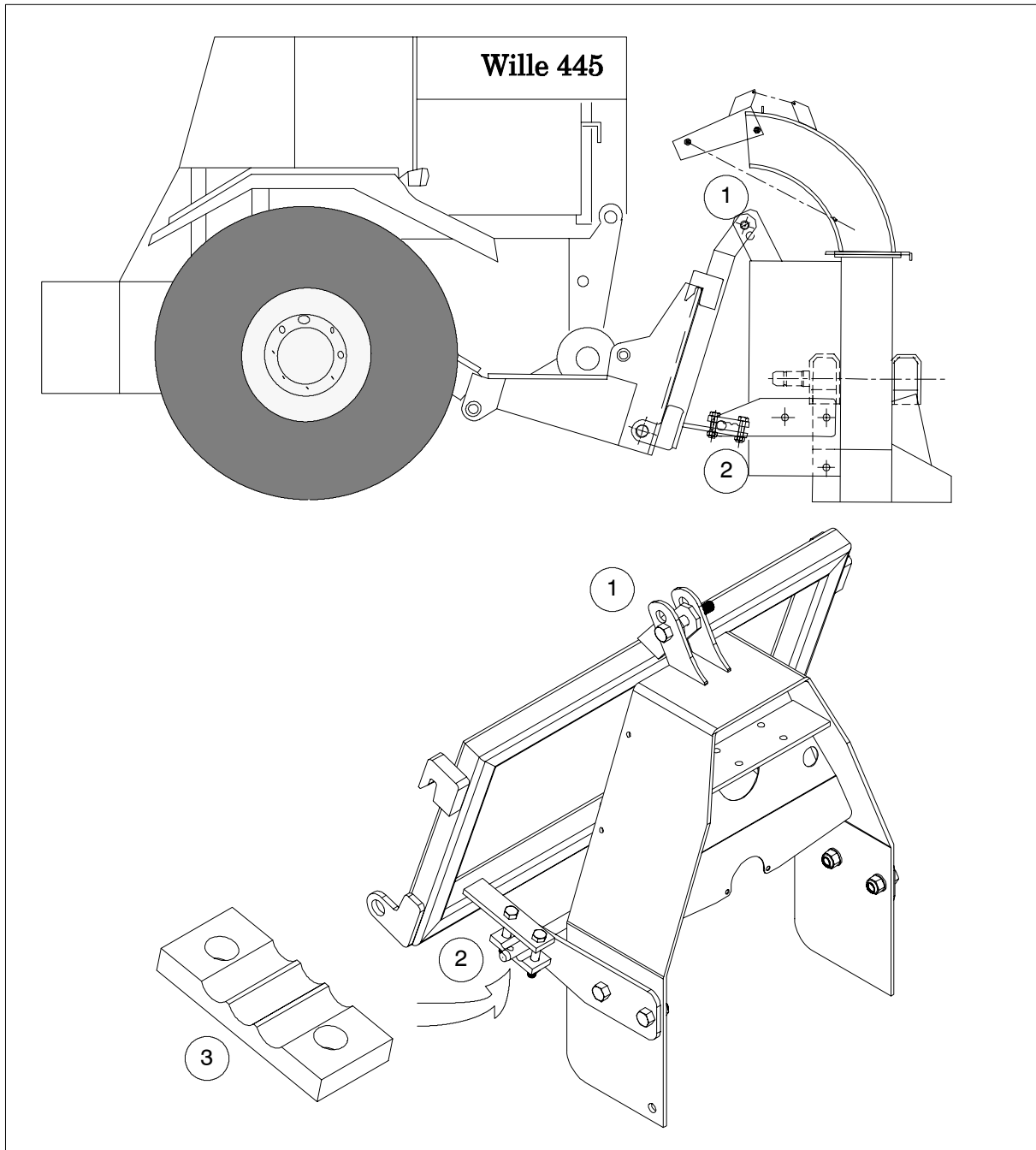


Fig. 7.Mounting on the Wille 445 adapter.

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6.3.MOUNTING ON THE BOBCAT ADAPTER (Mounting kit 03510800)

- The mounting assembly replaces the drawbar assy on the chipper frame.
- Attach the mounting plate to the top link brackets of the chipper with M20 pin and nut (1).
- Attach the left and right bracket to the chipper with four bolts (M20) (2).
- Attach the upper beam with four bolts (M12) (3).
- Attach the upper beam to the mounting plate with four bolts (M12) (4).
- Connect the hoses with quick–couplers to the Bobcat. Note the flow directions (see next page).

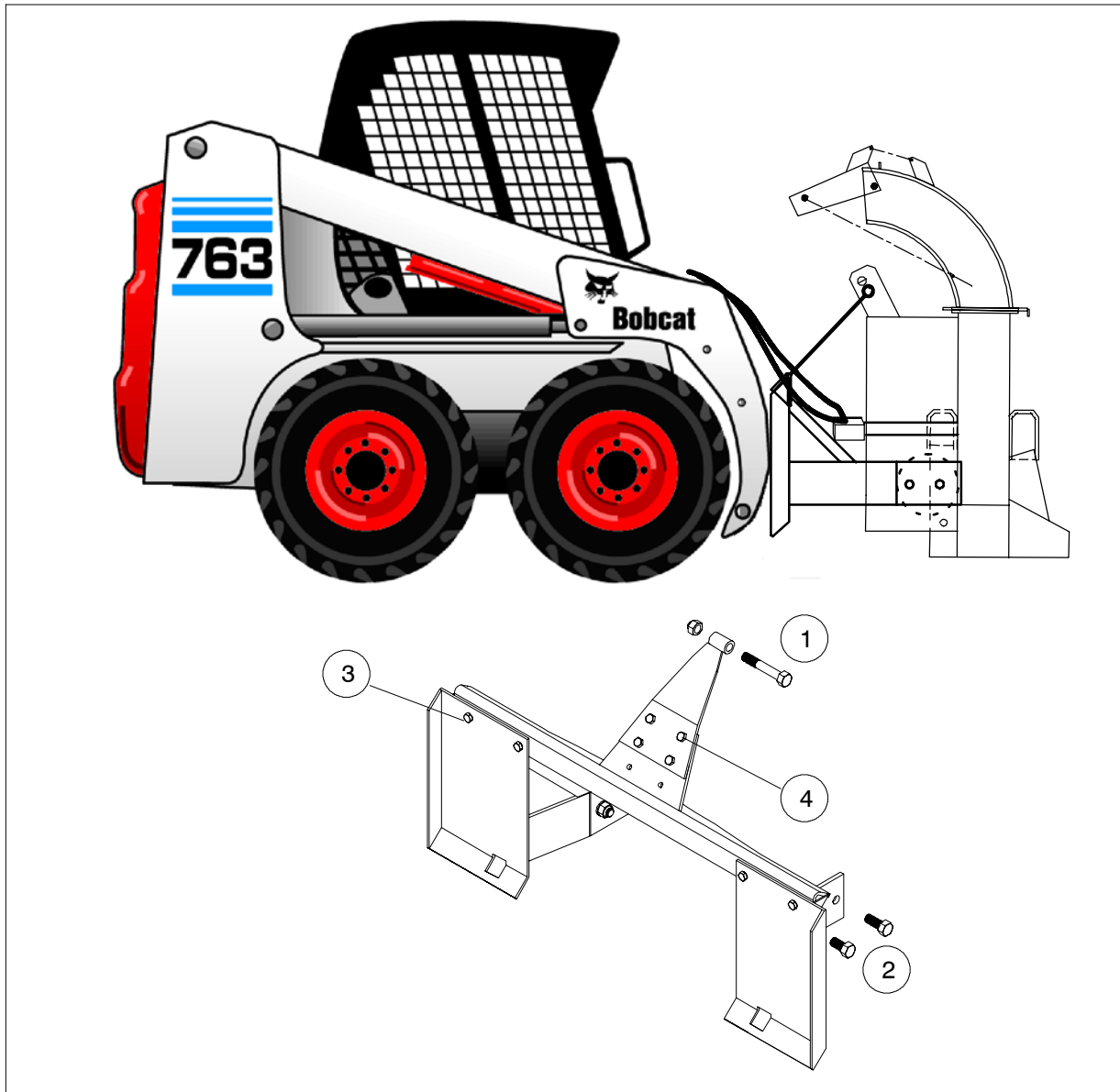


Fig. 8.Mounting on the Bobcat adapter.

6.4.INSTALLING THE HD 100

- The chipper can be driven with a separate hydraulic motor coupled to the chipper shaft.
1. Install the motor bracket on the chipper frame as shown in the figure.
 2. Attach the coupling sleeve with the keyway facing the hydraulic motor.
 3. Attach the hydraulic motor to the motor bracket. Check that the coupling sleeve is on the chipper's splined shaft.
 4. Connect the hydraulic hoses to the hydraulics of the drive machine.

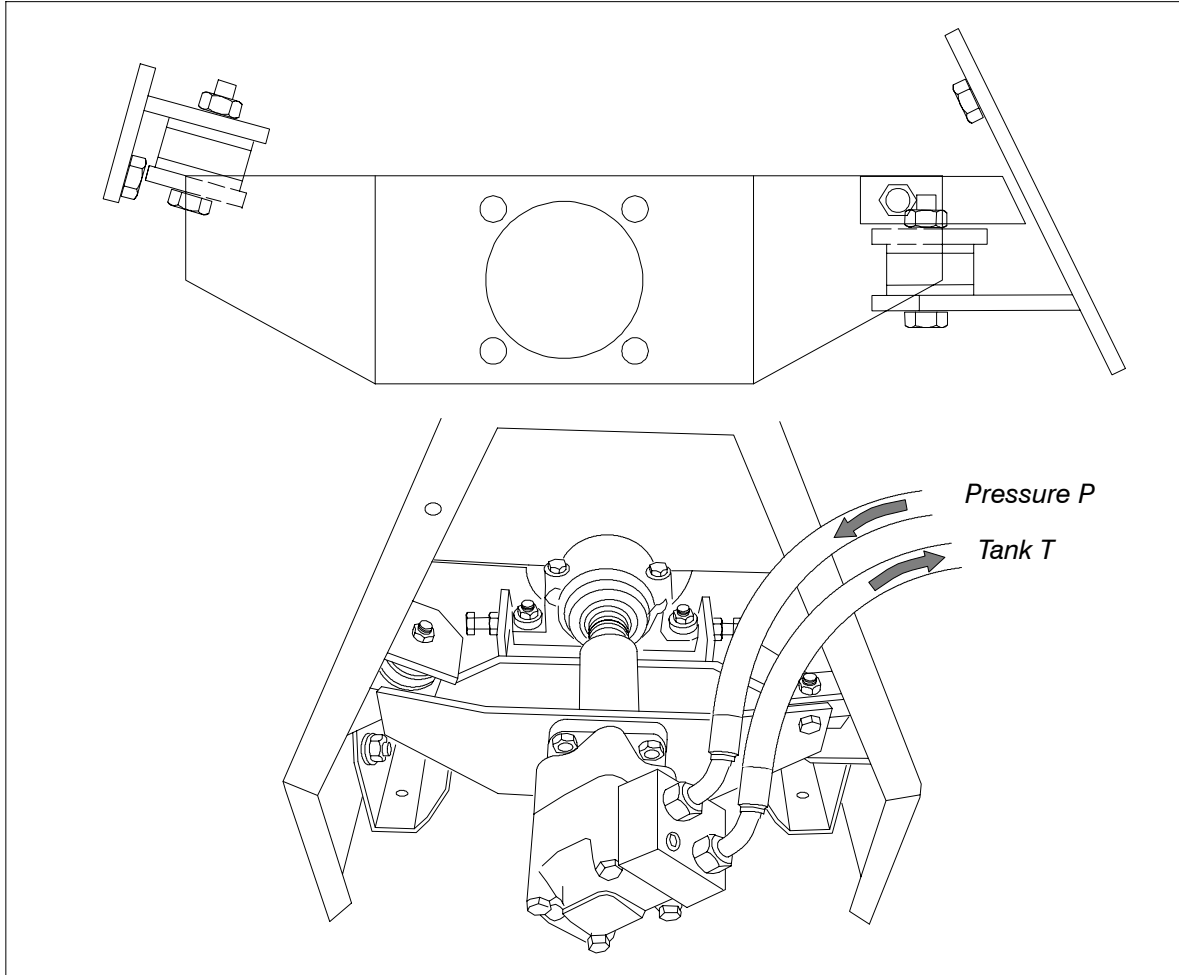


Fig. 9. Installing a separate hydraulic motor.

6.5. INSTALLING THE 160T $i=2.1$ BELT TRANSMISSION

1. Remove the PTO guard and install to the cover plate (1). Remove the mounting bolts of the small cover plate and remove the cover plate (2) from the mounting frame. These components will not be needed later. See Figure 10.
2. Attach the upper belt pulley assembly (3) and the regulator plate (4) to the chipper with the M12 bolts (5). Tighten the bolts lightly so that the bearing housing can be moved. See Figure 11.
3. Install the belt tightener (6) with the pin (7) and the adjusting assembly with the pin with hole (8). Secure the pins with cotter pins.
4. Install the belts on the upper belt pulley.
5. Push the lower belt pulley (9) onto the splined shaft without the tightening cone (10) while placing the belts into the grooves of the lower belt pulley.
6. Install the lower belt pulley tightening cone, NordLock lock washer pairs, and tightening cone bolts (M12) (11). Tighten lightly so that the lower belt pulley can be moved on the splined shaft.
7. Check the alignment of the belts – see Figure 12. Adjust the alignment of the belt pulleys by moving the upper belt pulley bearing housing and the lower belt pulley.
8. Tighten the bolts of the upper belt pulley bearing housing and the bolts of the lower belt pulley tightening cone (5 and 11) M12 to 80 Nm.
9. Re-check the alignment of the belts and adjust if necessary.
10. Adjust the belt tightener so that the roller touches the belts lightly, then tighten the adjusting bolt (12) by two to three turns. See Figure 13.

Be careful not to over-tighten, as this could damage the belts!

11. Attach the plate (1) to the mounting frame with bolts M8 (13).
12. Mount the chipper on the tractor. To check that everything is in order, run the chipper for a while at slow speed.
13. The belts must be run in, first without load for about $\frac{1}{2}$ hour and then with the chipper loaded lightly for the first 8 hours of actual use.
14. Observe the tightness of the belts and, if necessary, adjust with the adjusting bolt (12).

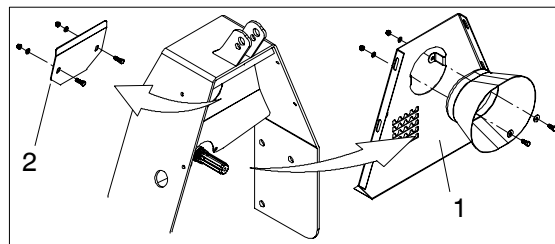


Fig 10. Removing the covers.

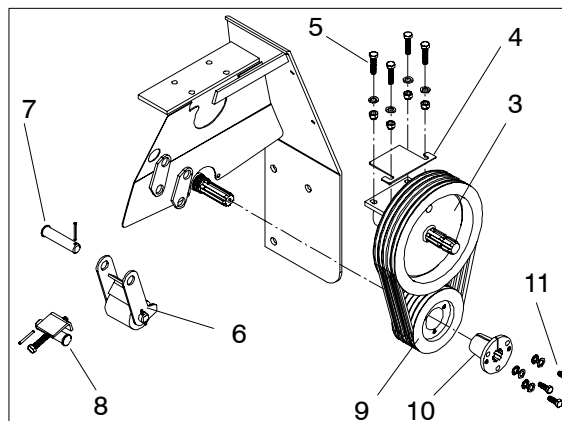


Fig 11. Installing the 160T belt transmission.

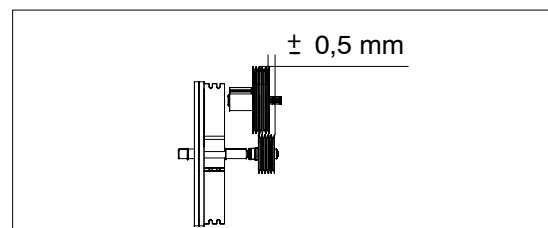


Fig 12. Belt alignment.

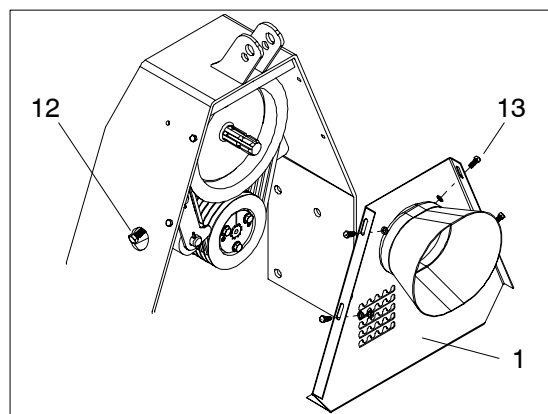


Fig 13. Installing the 160T belt transmission cover plate.

7. OPERATING THE CHIPPER

7.1. INSPECTIONS PRIOR TO OPERATION

- Park the chipper on level, hard ground.
- Stop the tractor's engine and ensure that the chipper is fully stopped before carrying out the inspections.
- Check that the disk rotates freely by turning the shaft, and ensure that there are no foreign objects inside the chipper.
- Ensure that all protective devices are intact and in place. Removal of the covers is prohibited.
- Direct the discharge pipe so that flying chips do not pose a risk to the operator or anyone else.

7.2. STARTING THE CHIPPER

- Start the chipper at low tractor engine speed and increase the speed slowly to the required chipping speed (540/1,000 rpm).
- The chipper is now ready for chipping.

7.3. STOPPING A TRACTOR-DRIVEN CHIPPER

- Slow the tractor engine speed to idle before disengaging the PTO. This is especially important with tractors featuring a PTO brake (e.g., Ford). Turn the PTO control lever slowly to the OFF position.

7.4. STOPPING A HYDRAULIC MOTOR DRIVEN CHIPPER

- **IMPORTANT!** When the chipper is driven by the HD100, it is extremely important to slow the driving engine speed to idle before disconnecting the hydraulics, **to prevent cavitation.**



Never disconnect the hydraulics from high rpm. This will cause a risk of cavitation and may damage the hydraulic motor. Check that no sounds indicating cavitation can be heard from the hydraulic motor.



After stopping the chipper, wait for all movement to stop. The disk continues rotating like a flywheel after the PTO is disengaged.

7.5. CHIPPING

- Before feeding in the material to be chipped, ensure that the wood is free from nails, stones, etc.
- Feed the wood from a standing position on the left side of the feeder.
- Push the wood inside the feed chute until chipping starts. Let go of the wood as soon as the self-feeding starts.
- If you are using a hydraulic feeder, refer to the separate chipping instructions in the feeder manual.



Do not use the chipper in temperatures below -20°C. This is to avoid damage due to brittleness – especially in the knives – caused by the cold. Avoid chipping wood that is frozen solid, as self-feeding is reduced in this case.



DANGER!

ROTATING KNIVES!

Knives can cause cutting injury. Do not reach inside the feed chute with hands or feet.



WARNING!

FIRE HAZARD!

Always keep adequate fire-fighting equipment on hand when using the chipper. Check the outside temperature of the chipper regularly. If the chipper heats up abnormally, stop the chipper and determine the cause of the overheating. Check the temperature of bearings regularly.

Pay special attention to careful maintenance, and keep the chipper free from dust.

If the chipper starts smoking, pour water down the feed chute.

7.6. REDUCING THE POWER DEMAND

When using a small tractor to power the CH160 chipper, select a PTO speed of 540 rpm. Run the tractor at full engine speed.

If the tractor's engine speed still decreases during chipping, reduce power demand by altering the chipper to cut with one knife.

The chipper is altered for single-knife operation simply by removing one knife and attaching it to the other side of the disk as a counterbalance, to retain the balance of the disk. Since the chipper is self-feeding, the feed speed of wood is also halved.

1. See Sections 9.1., "Removing the upper chamber" and 9.2., "Sharpening the knives".
2. Place a spring washer on the bolt. Turn the lock nut on the bolt and tighten to 45 Nm.
3. Place another spring washer on the bolt and turn the bolt through the knife into the disk. See Fig. 14. Tighten the bolts to 45 Nm. Attach the knife using all four bolts.
4. Interchange the cutting and counterbalance knives regularly to ensure the balance of the disk.

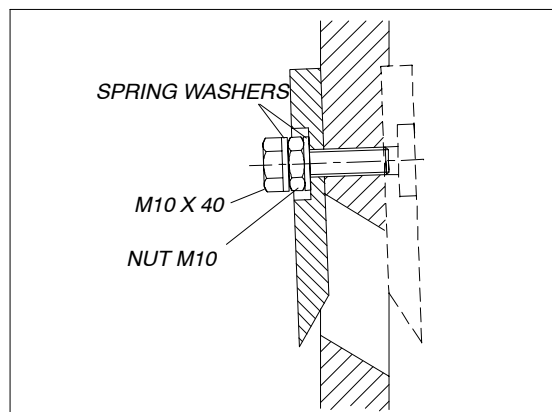


Fig. 14. Remove one knife and attach it to the other side of the disk. The dotted line shows the knife in its normal cutting position.

8.PERIODIC MAINTENANCE



Always disengage the PTO and stop the tractor and chipper fully before maintenance or repair work.

Ensure that the disk is fully stopped before inserting anything inside the feed chute.

Lock the disk before maintenance or repairs.

Park the chipper on hard, level ground to avoid it falling over.

Wear protective gloves when handling knives or anvils.

8.1.LUBRICATING THE BEARINGS

- The bearings are lubricated at the factory, and a similar lubricant should be used for subsequent lubrication (Shell Alvania Grease R 3. or Kendall L427). An excessive amount of grease causes overheating and impairs lubrication.
 - Lubricate the bearings every 200 working hours or at least once a year.
1. Open the upper bearing housing – see bearing housing assembly drawing, Fig. 28.. Remove old grease as carefully as possible and replace it with new grease. Do not fill the bearing housing with grease.
 2. Install the upper bearing housing and tighten to 50 Nm.

8.2.LUBRICATING THE PTO SHAFT

- Lubricate the PTO shaft prior to operation and regularly, as shown in Fig. 15..
- Lubricate the inner surface of the PTO shaft, accessed via the outer profile tube.
- Lubricate the shield tubes in wintertime to prevent them from freezing and sticking.

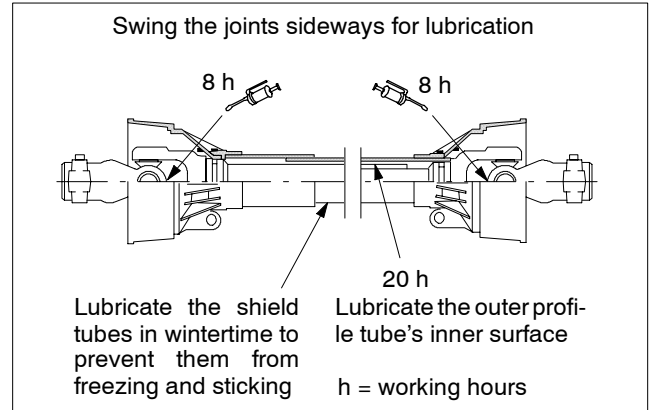


Fig. 15.Lubrication points and intervals for the PTO shaft.

8.3.PERIODIC INSPECTIONS

- With new machines, check the mounting bolts for tightness after the first operating hour, tightening them if necessary. Tightening torques are shown in table 2.
- Check the mounting bolts for tightness once a week.
- The knife-to-anvil clearance is adjusted to the specified values. For instructions on adjusting the clearance, see Section 9.7.

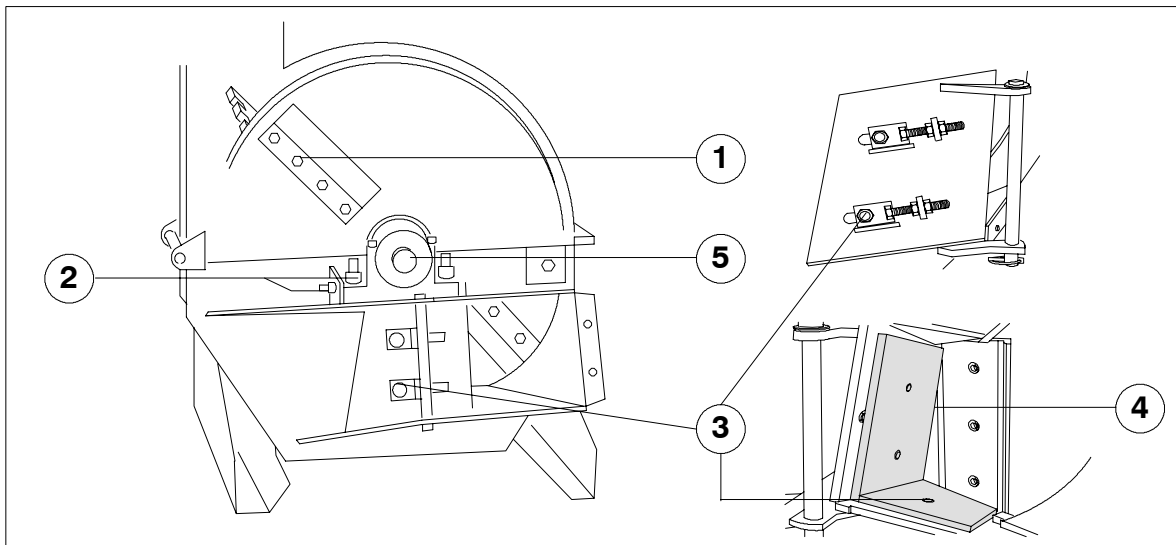


Fig. 16.Checklist for tightening and checking clearances.

Item	Width across flats, mm (inches)	Tightening torque, Nm (lbf)
1. Check the knife bolts for tightness.	17 mm (11/16")	60 (44 ³ / ₁₆)
2. Check the bearing housing bolts for tightness on both sides.	19 mm (3/4")	80 (55 ¹⁵ / ₁₆)
3. Check the anvil bolts for tightness.	24 mm (15/16")	200 (147 ¹⁵ / ₁₆)
4. Check the clearance between knives and vertical anvils.	19 mm (3/4")	1.2–1.5 mm (0.05–0.06")
5. Check the bearings for radial clearance.		0.02–0.03 mm (0.0008–0.0012")

Table 2.Tightening torques and clearances.

9. KNIFE AND ANVIL MAINTENANCE



DANGER!

Read the safety instructions. The disk continues rotating like a flywheel after the PTO is disengaged.



CAUTION!

Wear protective gloves when handling knives or anvils.

9.1. OPENING AND REMOVING THE UPPER CHAMBER

- Remove the upper chamber (A) securing bolts (M6) (B) and fastening bolts (M12) (C). Turn the upper chamber to the side.
- To remove the upper chamber, remove the securing bolt (M6) (D) and hinge bolt (M12) (E).
- Lock the disk with the lock bolt Fig. 18.
- Remove the feeder or turn it to the side.

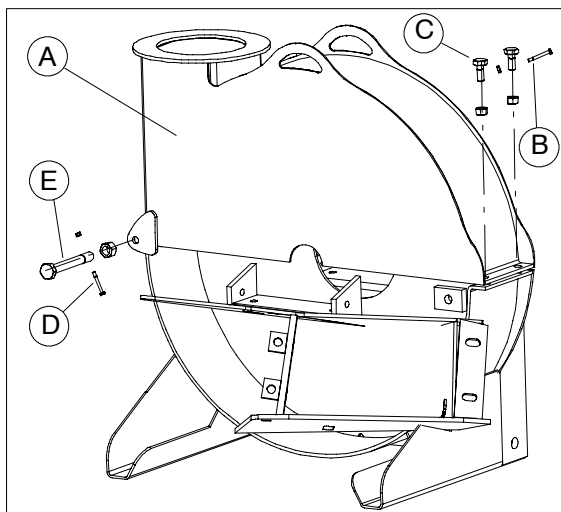


Fig. 17. Removing the upper chamber.

9.2. REMOVING THE KNIVES

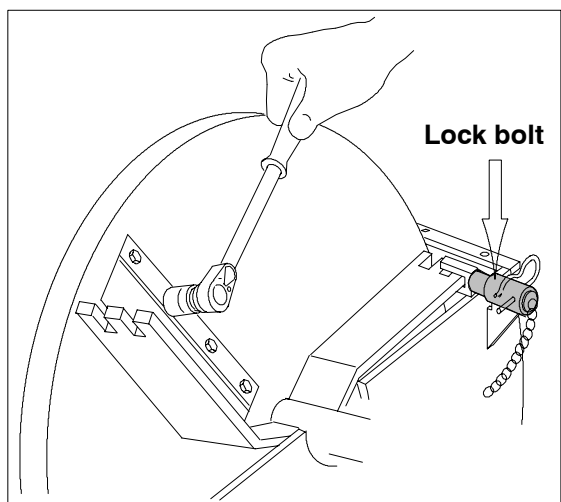


Fig. 18. Locking the disk and removing the lock nuts.

1. Remove the knife lock nuts (M10). Fig. 18.

2. Remove the knife fastening bolts (M10). Turn the wrench in such a way that your hands would not hit the knife if the wrench should slip. Fig. 19.

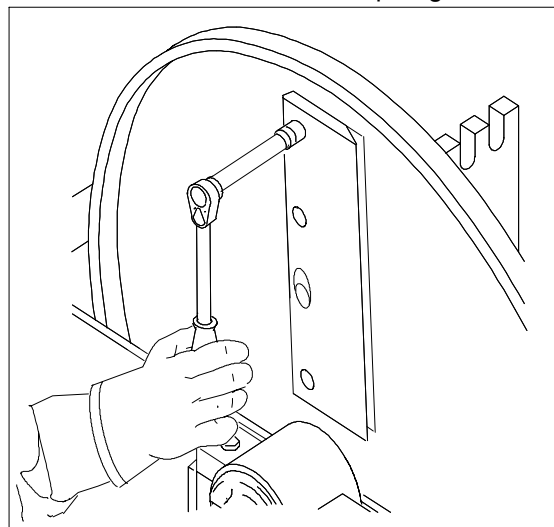


Fig. 19. Removing the knife fastening bolts.

9.3. SHARPENING THE KNIVES



CAUTION!

Sharpen all knives equally. This ensures disk balance.

Wear protective gloves when handling or anvils.

Avoid heating the knife during sharpening.

The knives need sharpening when:

- The self-feeding of wood has decreased;
- The power demand has increased;
- The chip surface is rough.

Normally, the knives can be sharpened several times without actually being removed (with, e.g., a sharpening stone or belt grinder).

More thorough conditioning is carried out with a surface grinder, with the knives removed.

The new knives are sharpened to a concave shape, $R=200$. The sharpening angle is 30° and hone angle is 45° . The hone angle prevents the edge from breaking.

Fig. 20.

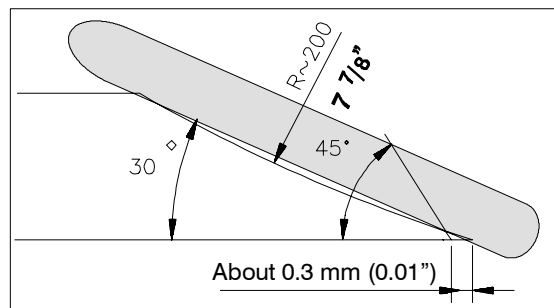


Fig. 20. The profile of a concave knife

It is recommended that the knives be sharpened to a concave shape. If this is not possible, the knife is sharpened to a flat profile. Fig. 21.

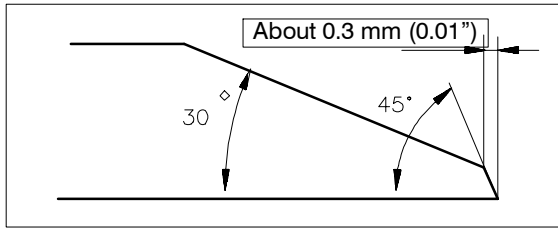


Fig. 21. A knife with a flat profile

The hone angle is ground to a 45° angle with two to three longitudinal strokes, using a level sharpening stone. Burrs are removed from the knife fastening bolt side, grinding with the surface.

Fig. 22.

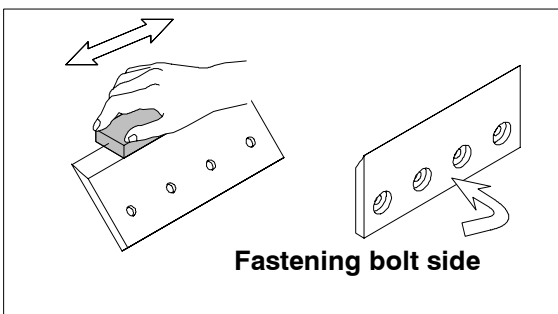


Fig. 22. Final grinding of the knife.

9.4.REMOVING THE ANVILS

The chipper features both a vertical and horizontal anvil. To remove the anvils, open the fastening bolts (A) and (B) (M16). The horizontal anvil fastening bolt (B) is located below the feed opening. Fig. 23.

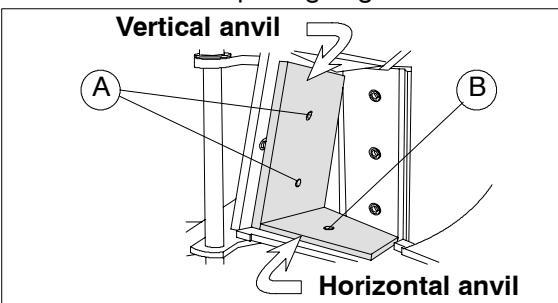


Fig. 23. Anvil fastening bolts

9.5.SHARPENING THE ANVILS

If you notice wear or rounding of the inner edge of the anvil, sharpen the anvils so that the original angles are retained. Fig. 24.

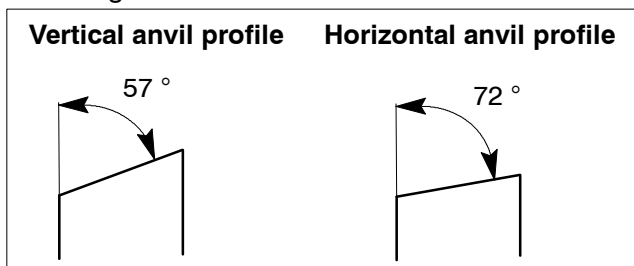


Fig. 24. Anvil profiles.

9.6.INSTALLING THE KNIVES AND ANVILS

- Check the condition of the fastening bolts and nuts.
- Install the knives and anvils and tighten the fastening bolts to the torques specified in Table 2..
- Adjust the knife-to-anvil clearance. For more information on adjusting the knife-to-anvil clearance, see Section 9.7.

9.7.ADJUSTING THE KNIFE-TO-ANVIL CLEARANCE

The need for adjusting the anvils is determined by the amount the knives are sharpened. Always check and, if necessary, adjust the clearance between knives and anvils:

- After a heavy sharpening;
- If the knives were removed – for example, due to sharpening;
- If new knives are replaced.

Check the clearance with a feeler gauge.

1. Loosen the anvil fastening bolts (A) and (B) (M16). Fig. 23.
2. Turn the disk so that a knife and the anvil are aligned. Place a feeler gauge between the knife and anvil. Adjust the vertical anvil clearance with the adjusting nuts (M12) (C, Fig. 25.) to 1.2–1.5 mm (0.05–0.06").
3. Tighten the adjusting nuts (C) and anvil fastening bolts (A).
4. Adjust the horizontal anvil clearance at the front edge to 4–5 mm (0.16 – 0.20").
5. Tighten the fastening bolt (B).
6. Re-check the knife-to-anvil clearances.

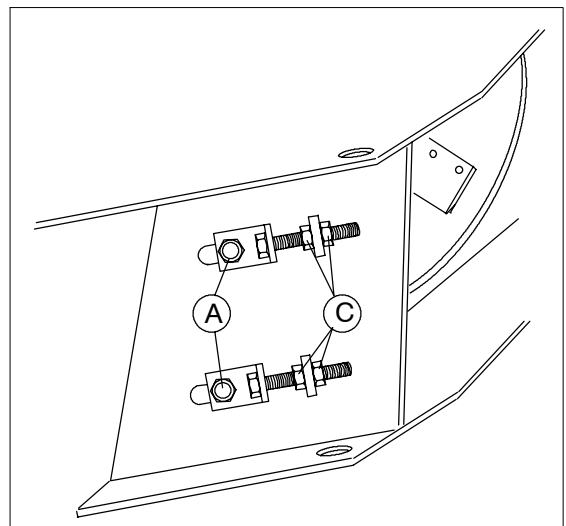


Fig. 25. Vertical anvil fastening and adjustment.

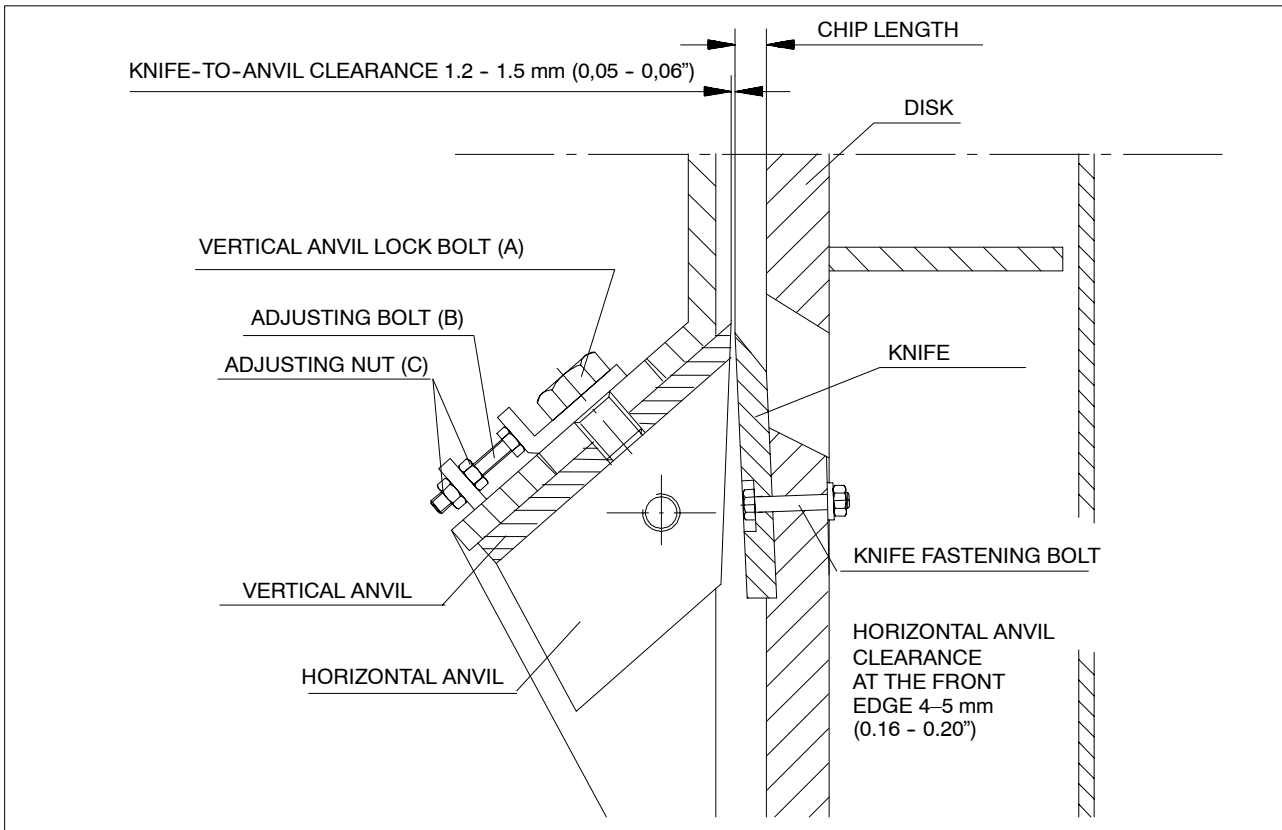


Fig. 26. Cross-section of the disk and knives/anvils.

10. REPLACING THE SPLINED SHAFT

1. Remove the upper chamber.
2. Remove the bearings (see Chapter 13). Welding damages the bearings.
3. Lift the disk up using a hoist.
4. Grind an 11-mm-deep groove about 17 mm from the edge of the shaft. See Fig. 27.
5. Heat the joint area, if necessary.
6. Move the splined shaft so that it comes loose and can be removed.
7. Clean the shaft hole and install the new shaft as shown in Fig. 27..
8. Make a fillet weld up to the surface level with three runs. Use ESAB 68.81, OK 48, OK Femax 38.65, or equivalent filler.

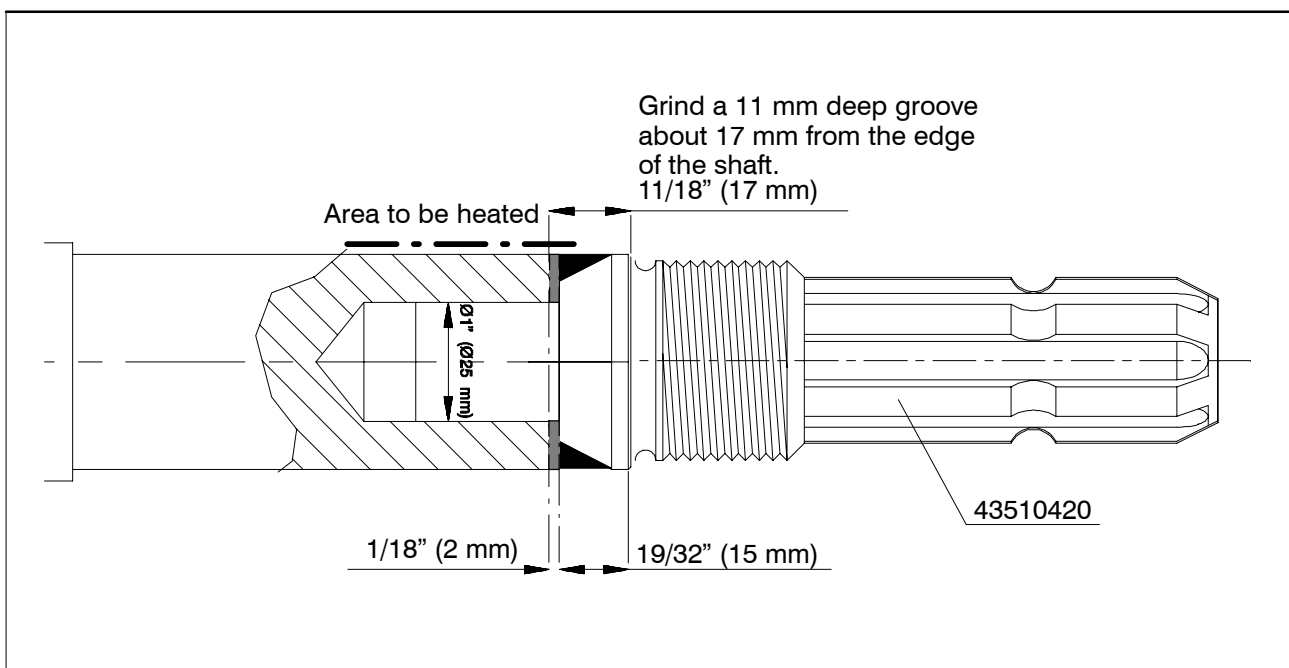


Fig. 27. Replacing the splined shaft.

11. REPLACING THE BEARING

1. Open the upper chamber. See Section 9.1.
2. Remove the fastening bolts M10 (7) and the upper bearing housings (6).
3. Mark the location of the tightening cone on the shaft.
4. Lift the disk.
5. Bend the claw of the securing ring (3) out from the notch on the axle nut and open the axle nut.
6. Remove the axle nut, securing ring, spacer ring (4), bearing (5), spacer ring (9), and tightening cone (10).
7. Install the tightening cone, spacer ring (9), and bearing on the disk shaft. Note the thickness of the spacer ring (9) – see Fig. 28.
8. Install the securing ring with the claws facing outwards and the inside claw in the groove of the tightening cone, and install the axle nut.
9. The inner ring of the bearing should press tightly against the tightening cone.
10. Tighten the axle nut with a hook spanner until the bearing is tightly on the cone, or to 80 Nm. However, the outer ring of the bearing should turn freely. Note the location of the tightening cone on the shaft.
11. Bend one claw of the securing ring (3) into a notch on the axle nut.
12. Install the other half of the dust cover and end plate on the bearing housing. Install the spacer ring (4) on the shaft.
13. Lower the disk to the bearing housing.
14. Attach the other half of the dust cover to the upper bearing housing; install the upper bearing housing fastening bolts and tighten to 50 Nm.
15. Lubricate the bearing housing. An excessive amount of grease causes overheating and impairs lubrication.

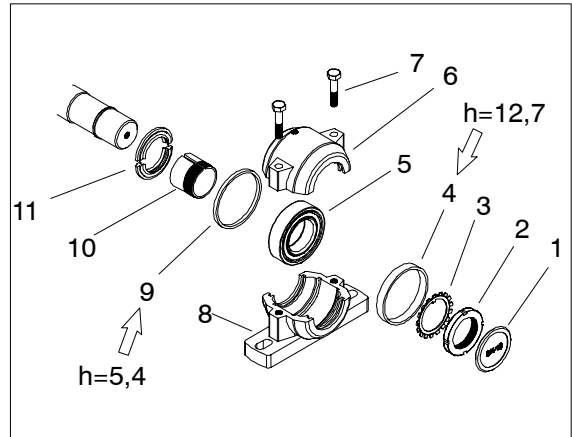


Fig. 28. Bearing, feeder side.

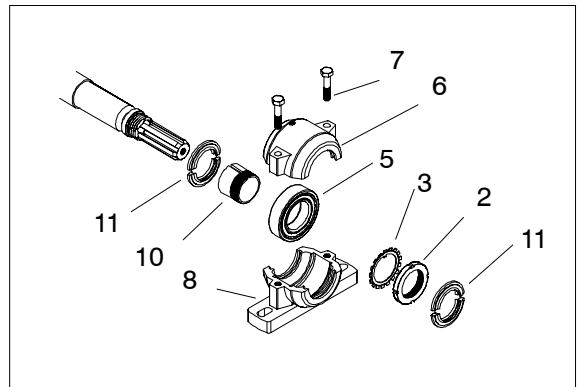


Fig. 29. Bearing, splined shaft side.

- The bearings at the feeder side and splined shaft side are different from each other. The bearing at the splined shaft side, Fig. 29., does not have spacer rings (4 and 9) and an end plate (1). The bearing at the splined shaft side has dust covers on both sides (11).

12. ADJUSTING THE BEARING CLEARANCE

1. Open the upper chamber. See Section 9.1.
2. Remove the fastening bolts (M10) (7) and the upper bearing housings (6).
3. Remove the grease from the bearing housing.
4. Lift the disk.
5. Bend the claw of the securing ring (3) out from the notch on the axle nut and open the axle nut.
6. Remove the spacer ring (3) if the bearing is at the feeder side.
7. Measure the radial clearance on top of the bearing between the rollers and outer ring with a feeler gauge. The clearance should be 0.02–0.03 mm (0.008–0.012"). Measure the clearance by pushing the feeler gauge between the rollers (point A, Fig. 31.) through the bearing and then moving the gauge back and forth between the rollers and outer ring – see Fig. 32. Do not force the feeler gauge through the clearance.
8. If the clearance exceeds 0.03 mm, bend the claw of the securing ring (2) out from the notch on the axle nut (1).
9. The bearing is tightened by turning the axle nut clockwise with a 70 mm (2 3/4") hook spanner until the right clearance is achieved. Do not tighten by hammering the axle nut.
10. Turn the axle nut clockwise until the notch is aligned with the nearest claw of the securing ring. Bend the claw into the notch. Do not bend the claw that was bent earlier.
11. Lower the disk to the bearing housing.
12. Install the upper bearing housing and tighten the bolts (6) to 50 Nm (36 7/8 lbf).
13. Lubricate the bearing housing. An excessive amount of grease causes overheating and impairs lubrication.

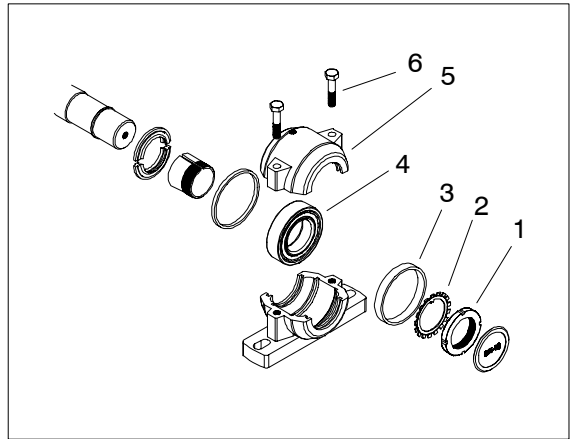


Fig. 30. Bearing, feeder side.

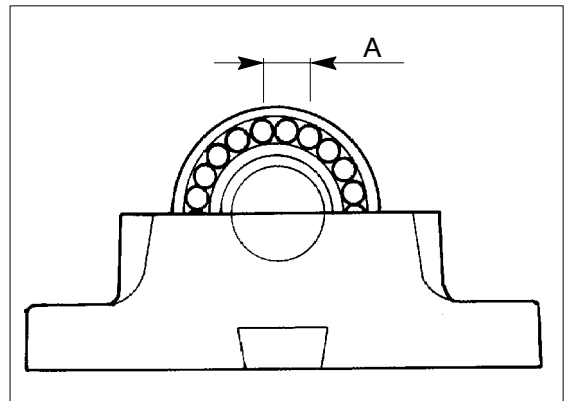


Fig. 31. Push the feeler gauge between the rollers and outer ring.

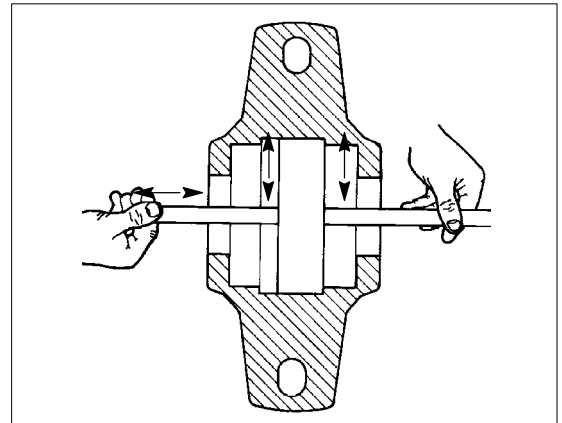


Fig. 32. Measuring the clearance.

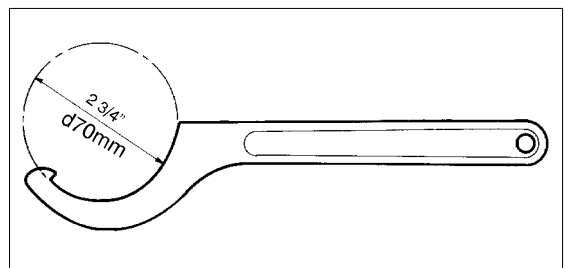
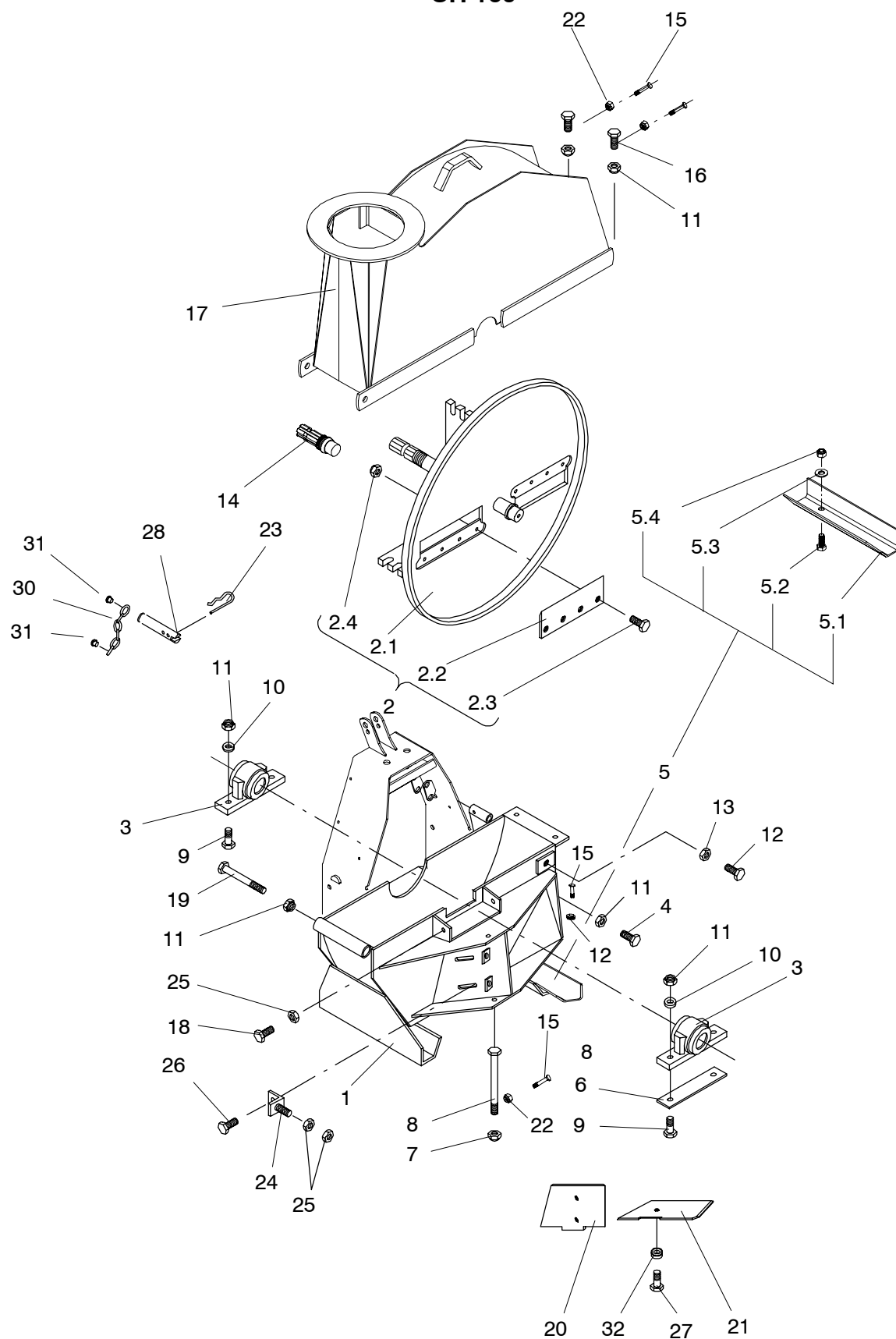


Fig. 33. Tightening the bearing.

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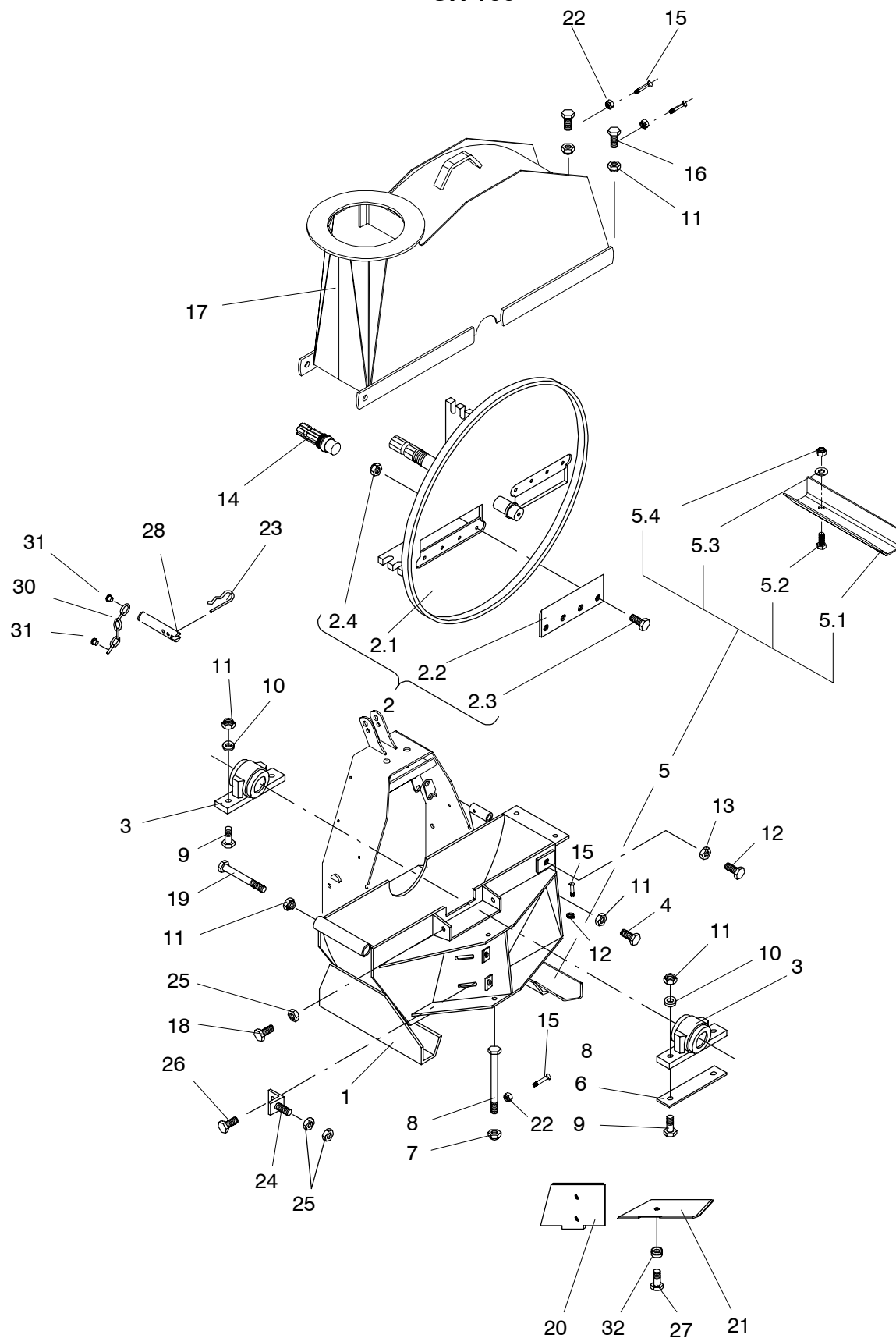
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1.	33514290 ...	Lower housing Unteres Gehäuse	Alarunko hitsaus Corps inférieur		1
2.	03510200 ...	Disc, complete Rotorscheibe, Komplet ...	Terälaikka koonta Disque, complet		1
2.1. ...	33510190 ...	Disc Rotorscheibe	Terälaikka Disque		1
2.2. ...	43510220 ...	Knife Messer	Terä Couteau		2
2.3. ...	52060258 ...	Hexagonal screw Sechskanntschaube	Kuusioruuvi Vis hexagonale	M10x40 DIN933	8
2.4. ...	52117108 ...	Lock screw Schlossmutter	Lukitusmutteri Ecrou de verrouillage	M10 DIN985	8
3.		Brearing Lager	Laakerointi Palier		
4.	43511630 ...	Safety screw Sicherheitbolzen	Turvaruuvi Cheville de sécurité	M12x50	2
5.	43513140 ...	Extension plate, mounting Platte, komple	Jalan jatke, asennus Rallonge—pied, complet		1
5.1. ...	43512490 ...	Extension plate Platte	Jalan jatke Rallonge—pied		1
5.2. ...	43512490 ...	Hexagonal screw Sechskanntschaube	Kuusioruuvi Vis hexagonale	M12x30 DIN933	1
5.3. ...	43341270 ...	Washer Scheibe	Aluslevy Rondelle	M12 DIN440	2
5.4. ...	52117124 ...	Lock nut Schlossmutter	Lukitusmutteri Ecrou de verrouillage	M12 DIN985	1
6.	43341270 ...	Mounting plate Montageplatte	Asennuslevy Plaque de montage		2
7.	52117165 ...	Lock nut Schlossmutter	Lukitusmutteri Ecrou de verrouillage	M16 DIN985	1
8.	43511660 ...	Safety screw Sicherheitbolzen	Turvaruuvi Cheville de sécurité	M16x240	1
9.	52062056 ...	Hexagonal screw Sechskanntschaube	Kuusioruuvi Vis hexagonale	M12x60 DIN931	4
10.	41215682 ...	Washer Scheibe	Aluslevy Rondelle		4
11.	52117124 ...	Lock nut Schlossmutter	Lukitusmutteri Ecrou de verrouillage	M12 DIN985	7
12.	52062916 ...	Hexagonal screw Sechskanntschaube	Kuusioruuvi Vis hexagonale	M16x60 DIN933	1
13.	52110079 ...	Nut Mutter	Kuusiomutteri Ecrou	M16 DIN934	1
14.	43510420 ...	Shaft Antriebswelle	Ura—akseli Arbre à nervures		1
15.	52060043 ...	Hexagonal screw Sechskanntschaube	Kuusioruuvi Vis hexagonale	M6x25 DIN933	5
16.	43511630 ...	Safety screw Sicherheitbolzen	Turvaruuvi Cheville de sécurité	M12x50	2
17.	33345150 ...	Upper housing Obere Gehäuse	Ylärunko hitsaus Corps supérieur		1

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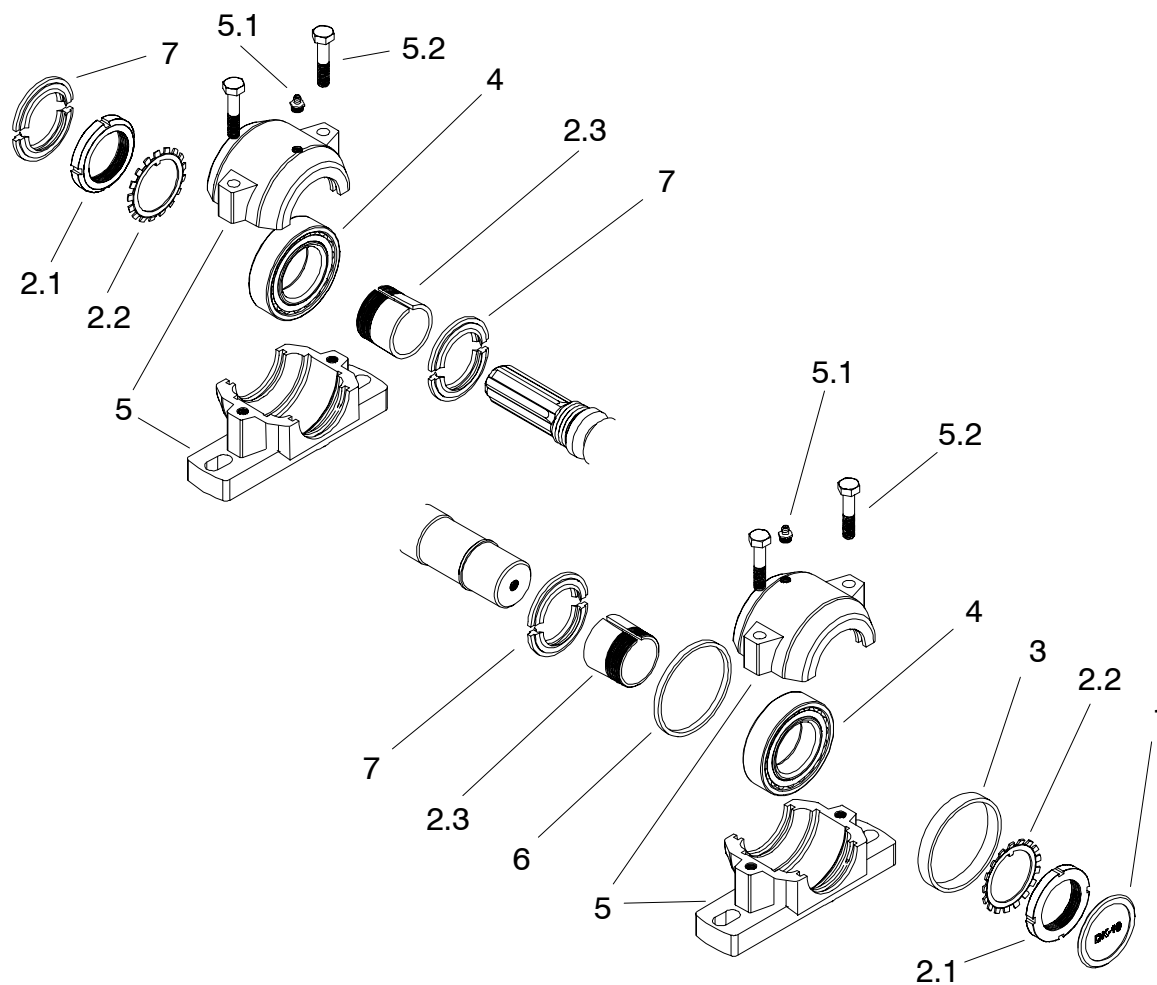
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18.	52062031 ...	Hexagonal screw Sechskanntschrabe	Kuusioruuvi Vis hexagonale	M12x40 DIN933	2
19.	43340546 ...	Screw Schraube	Ruuvi Vis		1
20.	33341009 ...	Vertical anvil Vertikaler Amboss	Pystyvastaterä Enclume verticale		1
21.	33340993 ...	Horizontal anvil Horizontales Amboss	Vaakavastaterä Enclume horizontale		1
22.	52117066 ...	Lock nut Schlossmutter	Lukitusmutteri Ecrou de verrouillage	M6 DIN985	5
23.	52842143 ...	Cotter pin Splint	Neulasokka Goupille	5x105	1
24.	43291103 ...	Anvil adjuster Ambossregler	Vastaterän säädin Régleur d'enclume		2
25.	52110053 ...	Nut Mutter	Kuusiomutteri Ecrou	M12 DIN934	8
26.	52062106 ...	Hexagonal screw Sechskanntschrabe	Kuusioruuvi Vis hexagonale	M16x30 DIN933	2
27.	52062098 ...	Hexagonal screw Sechskanntschrabe	Kuusioruuvi Vis hexagonale	M16x25 DIN933	1
28.	43341429 ...	Locking pin Schloßstift	Lukistustappi Doigt de verrouillage		1
29.	52211042 ...	Spring washer Federscheibe	Jousialuslevy Rondelle ressort	M10 DIN127	1
30.	03291143 ...	Chain Kette	Ketju Chaîne	250	1
31.	52832045 ...	Rivet Niete	Vetoniitti Rivet explosif	4,8x19 Fe	2
32.	43310499 ...	Washer Scheibe	Aluslevy Rondelle		1

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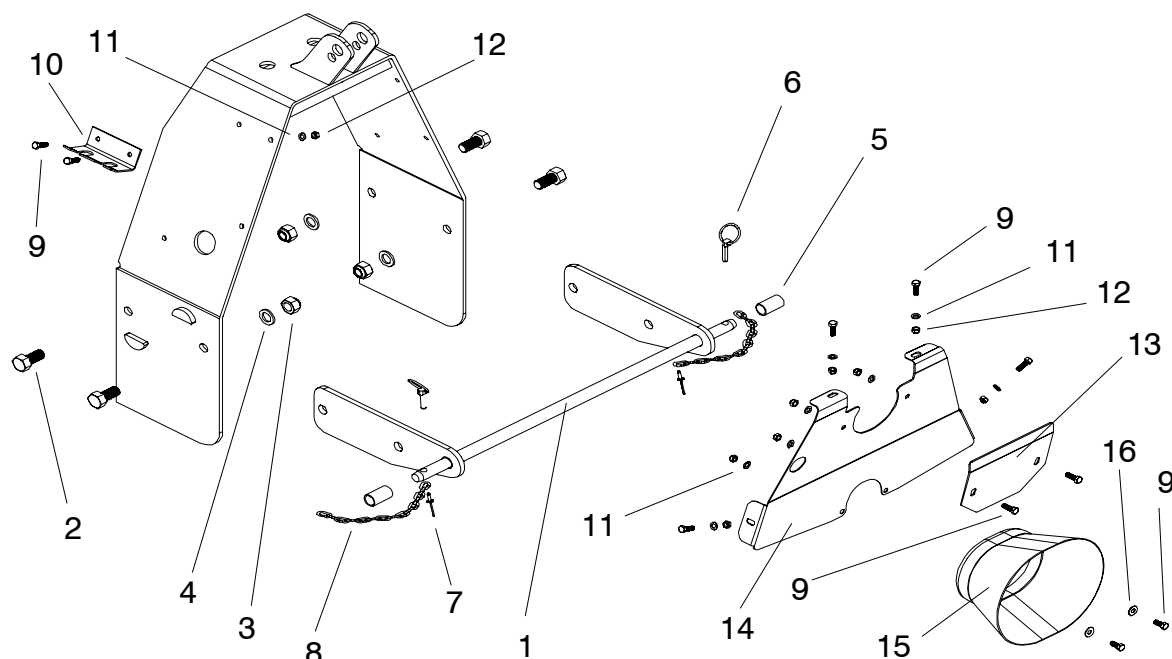
Bearing – Laakerointi – Lager – Palier



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1.	54513569 ...	End plate Abschlussplatte	Päätylevy Plaque d'extrémité		1
2.	54512371 ...	Tightener sleeve, complete Befestigungsbusche, Komplet	Kiristysholkkisarja Cône de serrage, complete		2
2.1.		Axle nut Achsemutter	Akselimutteri Garniture d'étanchéité		1
2.2.		Lock plate Schlossplatte	Lukkolevy L'anneau de calage		1
2.3.		Tightener sleeve Befestigungsbusche	Kiristysholkki Cône de serrage		1
3.	43513360 ...	Spacer ring Distanzring	Soviterengas Anneau intermédiaire	D90x12,7	1
4.	54512363 ...	Bearing Lager	Laakeri Palier		2
5.	54513590 ...	Bearing housing Lagergehäuse	Laakeripesä Corps de palier		2
5.1. ...	52401015 ...	Grease nipple Schmiernippel	Rasvanippa Graisseur	1/8"	1
5.2.		Hexagonal screw Sechskanntschraube	Kuusioruuvi Vis hexagonale	M10x50 DIN931 10.9	2
6.	43513350 ...	Spacer ring Distanzring	Soviterengas Anneau intermédiaire	D90x5,4	1
7.	52334232 ...	Seal Dichtung	Pölysuojapari Garniture d'étanchéité		1 / 2

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Mounting frame–Vetokolmio–Zugrahmen–Support de montage

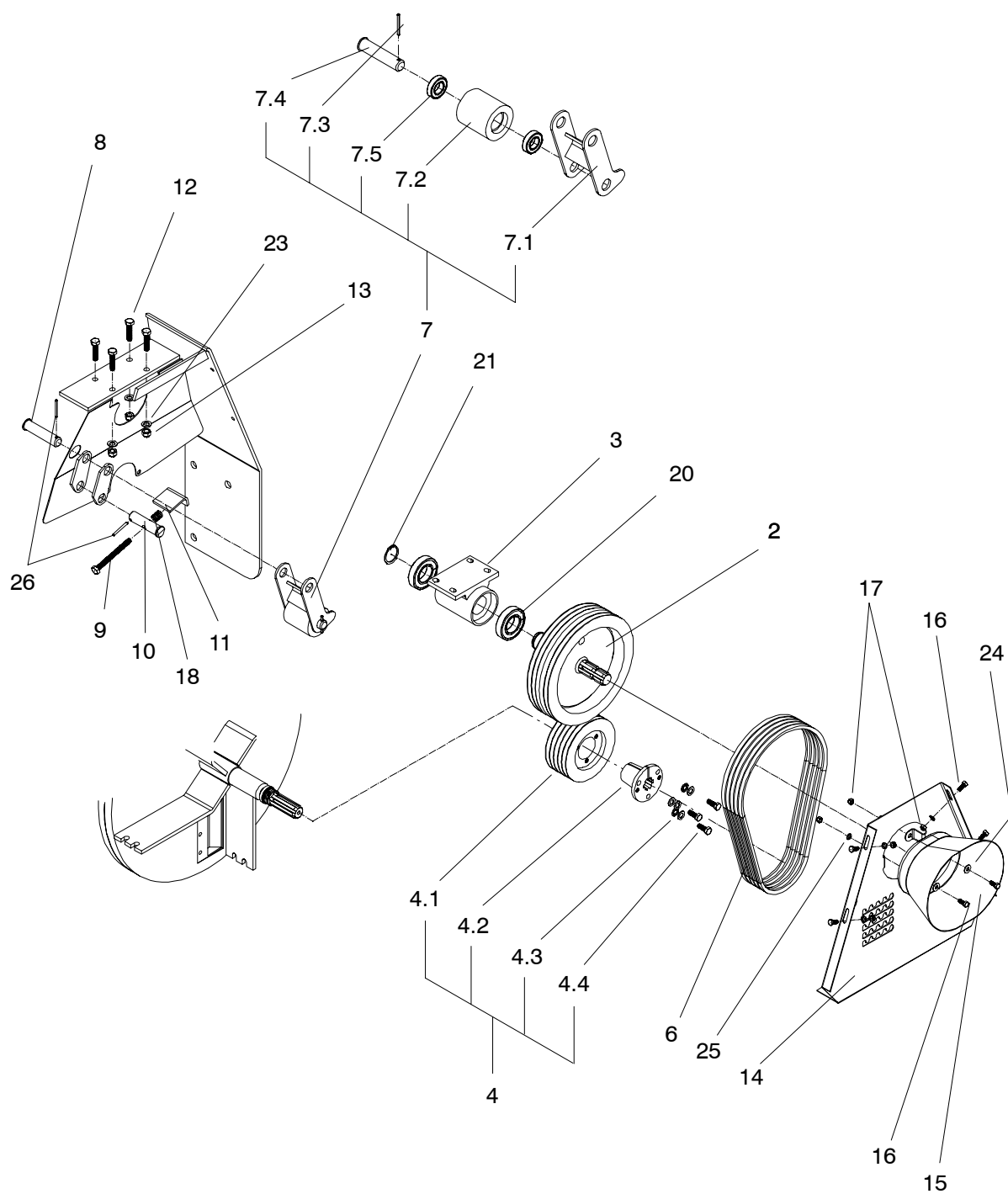


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1.	43343790	Draw bar Zugstange	Vetokarttu Barre d'attelage		1
2.	54512371	Hexagonal screw Sechskanntschraube	Kuusioruuvi Vis hexagonale	M20x50 DIN933	4
3.	52117207	Lock nut Schlossmutter	Lukitusmutteri Ecrou de verrouillage	M20 DIN985	4
4.	52211083	Spring washer Federscheibe	Jousialuslevy Rondelle ressort	M20 DIN127	4
5.	40293797	Sleeve Büsche	Bague Bague		2
6.	52842150	Ring cotter Ringsplint	Rengassokka Clavette ronde	10x45	2
7.	52832094	Rivet Niete	Vetoniitti Rivet explosif	4,8x15 Fe	2
8.	54826110	Chain Kette	Ketju Chaîne	300 mm	2
9.	52060175	Hexagonal nut Sechskanntschraube	Kuusioruuvi Vis hexagonale	M8x25 DIN933	10
10.	43482290	Hose fastener Montageplatte	Letkunpidike Support de flexible		1
11.	52200037	Washer Scheibe	Aluslevy Rondelle	M8 DIN126	10
12.	52117082	Lock nut Schlossmutter	Lukitusmutteri Ecrou de verrouillage	M8 DIN985	10
13.	43514240	Plate Platte	Levy Plat		1
14.	43514210	Plate Platte	Suojalevy Plat		1
15.	43511780	Cover of the universal shaft Schutz der Gelenkwelle	Nivelakselin suoja Couvercle d'arbre de prise de force		1
16.	52200334	Washer Scheibe	Aluslevy Rondelle	M8 DIN440	2

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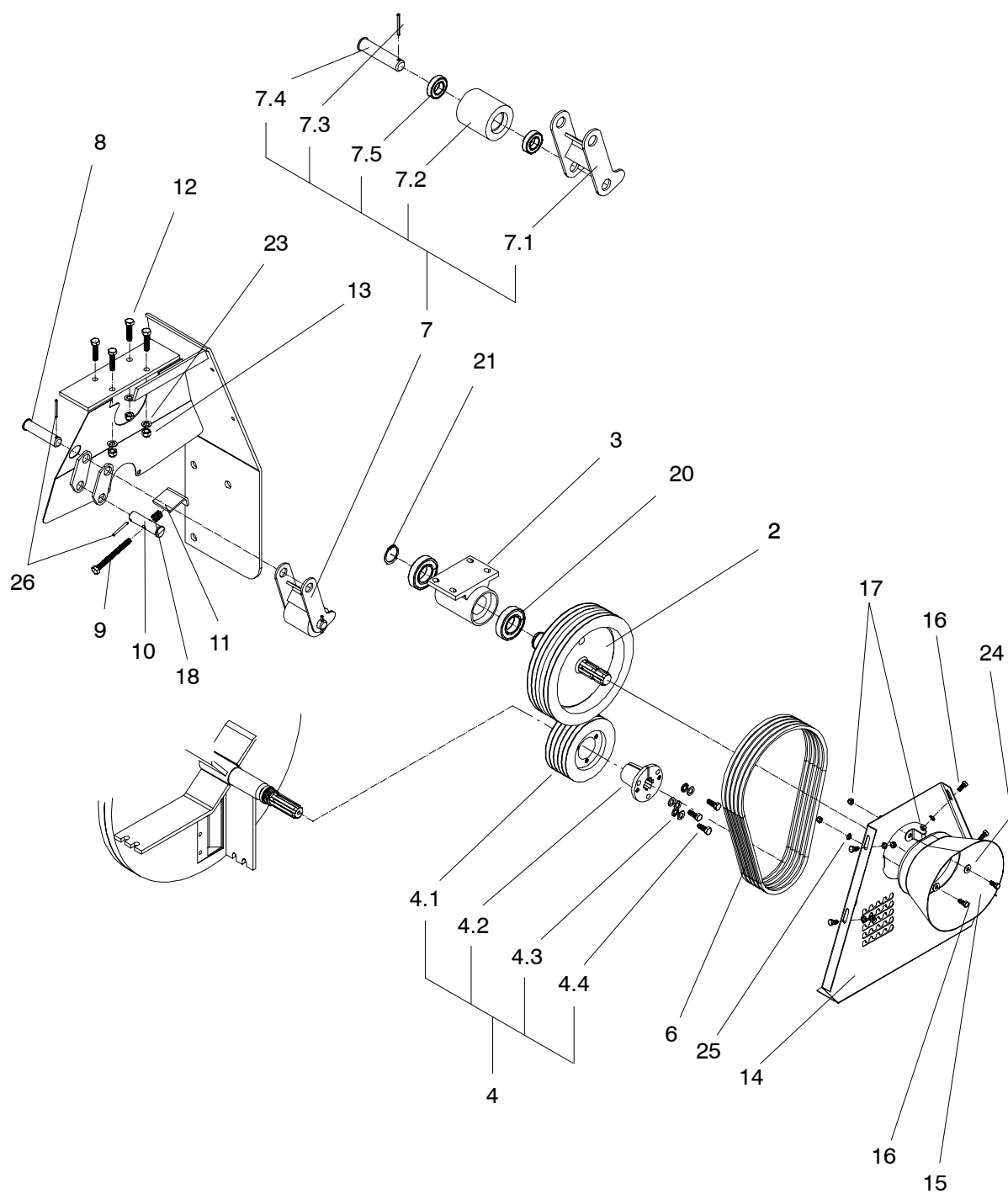
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	43343790 ...	Belt transmission, complet .. Riemenantriebs, Komplet	Hihnavaihde, täydellinen Transmission par courroies, complète		
1.	—				
2.	43514300 ...	Belt pulley Bandscheibe	Hihnapyörä Poulie à courroie	D315	1
3.	43514450 ...	Bearing housing Lagergehäuse	Laakeripesä Corps de palier		1
4.	43514330 ...	Belt pulley, assy Bandscheibe, Komplet	Hihnapyörä koonta Poulie à courroie, complète		1
4.1. ...	43514350 ...	Belt pulley Bandscheibe	Hihnapyörä Poulie à courroie	D150	1
4.2. ...	43514340 ...	Cone sleeve Spannkonus	Kartioholkki Manchon conique		1
4.3. ...	52214269 ...	Washer Scheibe	Aluslevy Rondelle	M12 Nord—lock	3
4.4. ...	52062023 ...	Hexagonal screw Sechskanntschaube	Kuusioruuvi Vis hexagonale	M12x30 DIN933	3
5.	—				
6.	54822382 ...	Belt Band	Kiilahihna Courroie		5
7.	43512020 ...	Belt tightener, complete Riemenspanner, Komplet	Kiristin, täydellinen Tendeur de courroie, complet		1
7.1. ...	43512030 ...	Belt tightener Riemenspanner	Hihnankiristin Tendeur de courroie		1
7.2. ...	43341106 ...	Roller Rolle	Rulla Galet		1
7.3. ...	52840055 ...	Cotter pin Scherensplint	Jousisokka Goupille fendue	5x50 DIN1481	1
7.4. ...	43341114 ...	Pin Zapfen	Tappi Cheville		1
7.5. ...	54511134 ...	Bearing Lager	Laakeri Palier		2
8.	43341114 ...	Pin Zapfen	Tappi Cheville		1
9.	52063658 ...	Hexagonal screw Sechskanntschaube	Kuusioruuvi Vis hexagonale	M12x120 DIN933 ...	1
10.	43511950 ...	Pin Zapfen	Tappi Cheville		1
11.	43512050 ...	Spring pusher Spring pusher	Jousen painin Poussoir de ressort		1
12.	52090560 ...	Hexagonal screw Sechskanntschaube	Kuusioruuvi Vis hexagonale	M12x50 DIN933	4
13.	52117124 ...	Lock nut Schlossmutter	Lukitusmutteri Ecrou de verrouillage	M12 DIN985	4
14.	43511970 ...	Plate Platte	Levy Plat		1
15.	43511780 ...	Cover of the universal shaft .. Schutz der Gelenkwelle	Nivelakselin suoja Couvercle d'arbre se prise de force		1
16.	52060126 ...	Hexagonal screw Sechskanntschaube	Kuusioruuvi Vis hexagonale	M8x20 DIN933	6
17.	52117082 ...	Lock nut Schlossmutter	Lukitusmutteri Ecrou de verrouillage	M8 DIN985	6
18.	43402150 ...	Spring Feder	Jousi Ressort	26x19x3	1

CH 160

160T i=2,1



03514390

CH 160

160T i=2,1

Part Osa Teil Pièce	Order No. Til.nro. Best Nt No. de ordre	Description Bezeichnung	Nimitys Désignation	Remarks Huom. Obs Obs	Qty. Kpl. St. Pc.
19.	—				
20.	54512140	Bearing Lager	Laakeri Palier		2
21.	52230257	Circlip Sicherungsring	Akselivarmistin Arrêt de sécurité de l'ax	45x2,5 DIN471	1
22.	—				
23.	52200466	Washer Scheibe	Aluslevy Rondelle	M12 DIN126	4
24.	52200334	Washer Scheibe	Aluslevy Rondelle	M8 DIN440	2
25.	52200037	Washer Scheibe	Aluslevy Rondelle	M8 DIN126	6
26.	52840055	Cotter pin Scherensplint	Jousisokka Goupille fendue	5x50 DIN1481	2

HUOM! Uusi 2,1 –välitys koneesta nro 351 1623 alkaen. Voit tarkistaa hihnapyörien koosta, kumpi välitys koneessa on:

AIKAISEMPI 1,7 –välitys: hihnapyörien ø 307 mm ja 178 mm

UUSI 2,1 –välitys: hihnapyörien ø 315 mm ja 150 mm

Hihnapyörä ø 178 mm voidaan korvata hihnapyörällä, jonka halkaisija on 150 mm. Tällöin tarvitaan lyhyemmät hihnat (tilausnro 54822580, SPA 1232).

NOTICE! New 2,1 belt transmission from machine serial nr. 351 1623. You will see by the size of the belt pulley, which transmission does the machine include:

PREVIOUS 1,7 belt transmission: belt pulleys ø 307 mm and 178 mm

CURRENT: 2,1 –belt transmission: belt pulleys ø 315 mm ja 150 mm

The belt pulley ø 178 mm can be replaced with belt pulley, which diameter is 150 mm. Then shorter belts are needed (order nr. 54822580, SPA 1232).

ACHTUNG! Von Seriennummer 351163 an ist die Keilriemenübersetzung ein neues Modell. Sie können die Übersetzung von den Grössen der Rädern identifizieren:

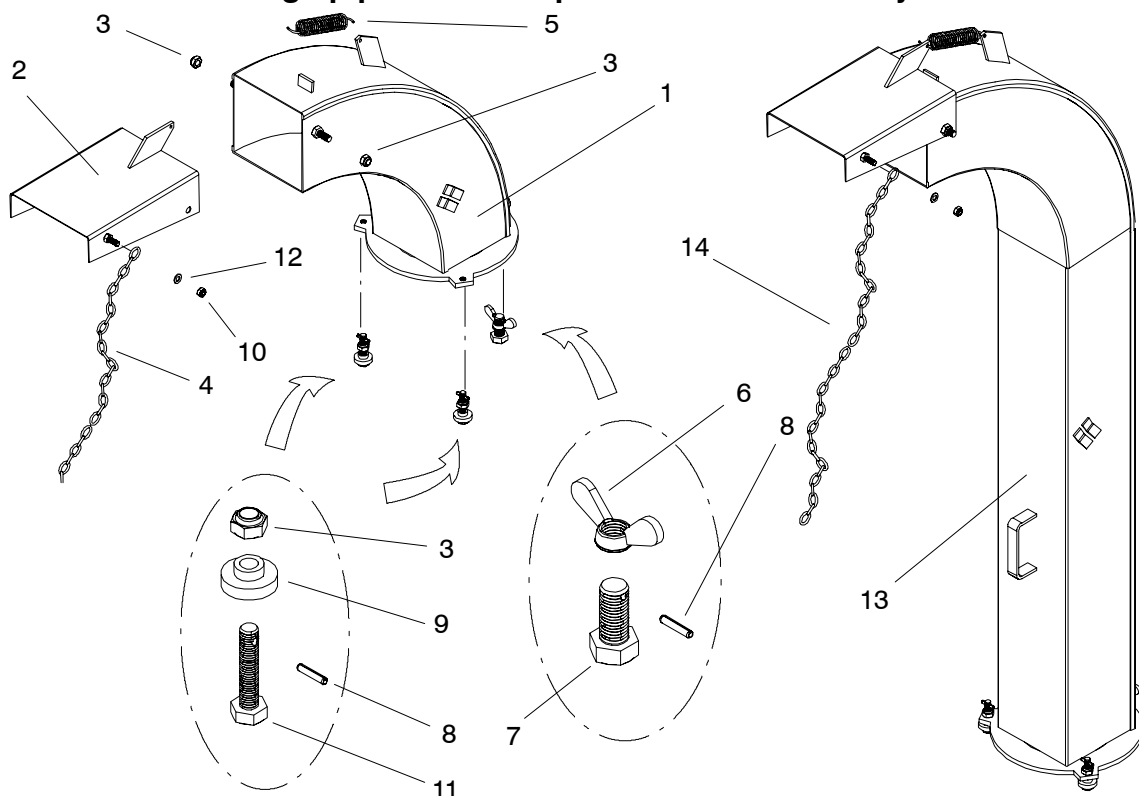
Mit früherer Übersetzung (i= 1,7) waren die Riemenräder wie folgt: ø 307 mm und ø 178 mm

Neue Übersetzung (i=2,1) ist sind die Riemenräder wie folgt: ø 315 mm und ø 150 mm

Das Rad ø 178 kann man durch das Rad ø 150 mm ersetzen. Dann braucht man kürzere Keilriemen (Bestellungsnummer 54822580, SPA 1232).

CH 160

Discharge pipe – Puhallusputki – Auswurfroh – Tuyau d'écoulement

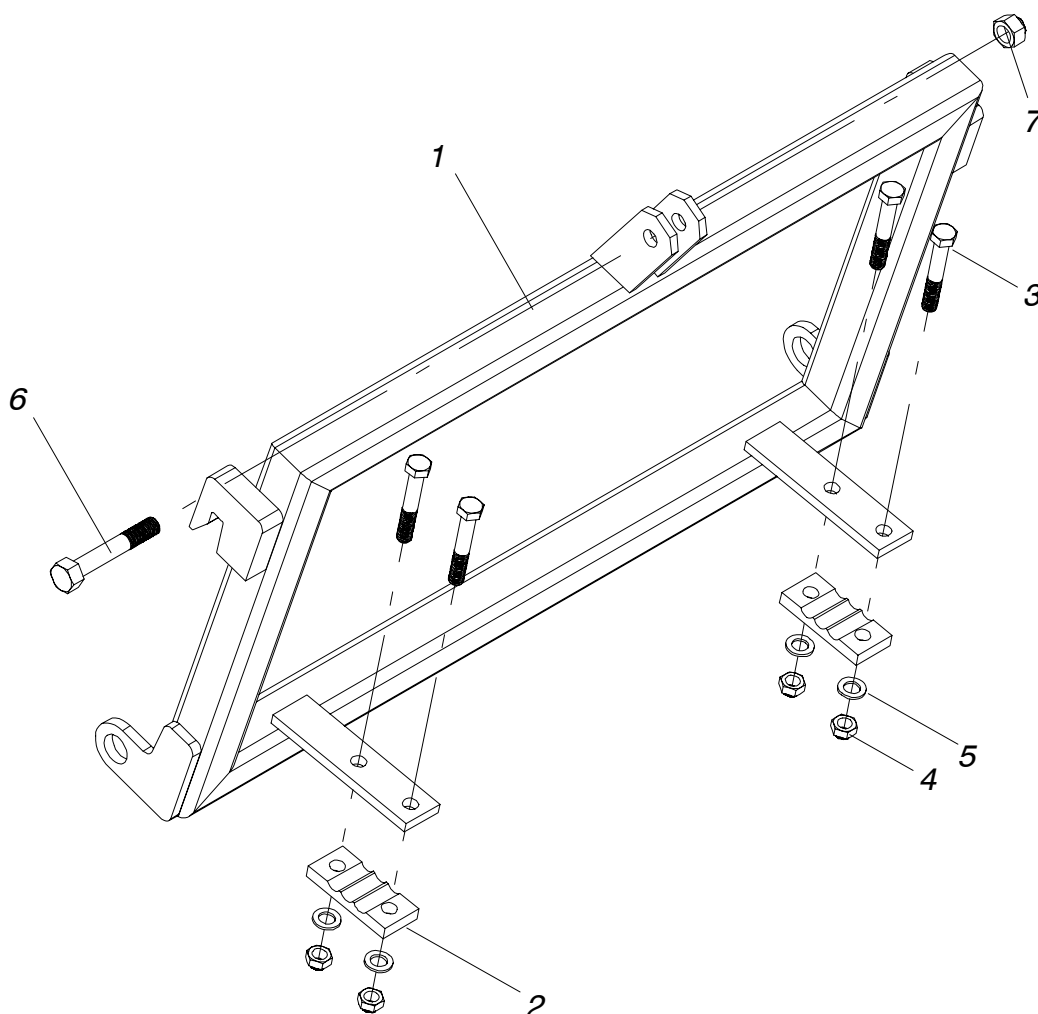


03514630 / 4640

Part Osa Teil Pièce	Order No. Til.nro. Best Nt No. de ordre	Description Bezeichnung	Nimitys Désignation	Remarks Huom. Obs Obs	Qty. Kpl. St. Pc.
	03514630	Short discharge pipe,complete Kurze Auswurfrohr,Komplet	Lyhyt puhallusputki,täydellinen Tuyau d'écoulement court,complete		
	03514640	Long discharge pipe,complete Lange Auswurfrohr, Komplet	Pitkä puhallusputki,täydellinen Tuyau d'écoulemen long,complete		
1.	43514600	Short discharge pipe Kurze Auswurfrohr	Lyhyt puhallusputki Tuyau d'écoulement court		1
2.	43510240	Lid Klappe	Lippa Couvercle		1
3.	52117108	Lock nut Schlossmutter	Lukitusmutteri Ecrou de verrouillage	M10 DIN985	4
4.	03510300	Chain Kette	Ketju Chaîne	600	1
5.	94612082	Spring Feder	Jousi Ressort	26x2,6x106	1
6.	43340769	Finger nut Schraubenmutter	Siipimutteri Ecrou à oreilles		1
7.	43511670	Safety screw Sicherheitbolzen	Turvaruuv Cheville de sécurité	M16x35	1
8.	52840295	Cotter pin Splint	Jousisokka Goupille	5x24 DIN1481	3
9.	43340751	Reel Rolle	Rulla Galet		2
10.	52117082	Lock nut Schlossmutter	Lukitusmutteri Ecrou de verrouillage	M8 DIN985	1
11.	43511650	Safety screw Sicherheitbolzen	Turvaruuv Cheville de sécurité	M10x50	2
12.	52200334	Washer Scheibe	Aluslevy Rondelle	M8 DIN440	1
13.	43514610	Long discharge pipe Lange Auswurfrohr	Pitkä puhallusputki Tuyau d'écoulement long		1
14.	03514590	Chain Kette	Ketju Chaîne	1100	1

CH 160

Adapter – Sovitus – Adaption – Adaptateur Wille

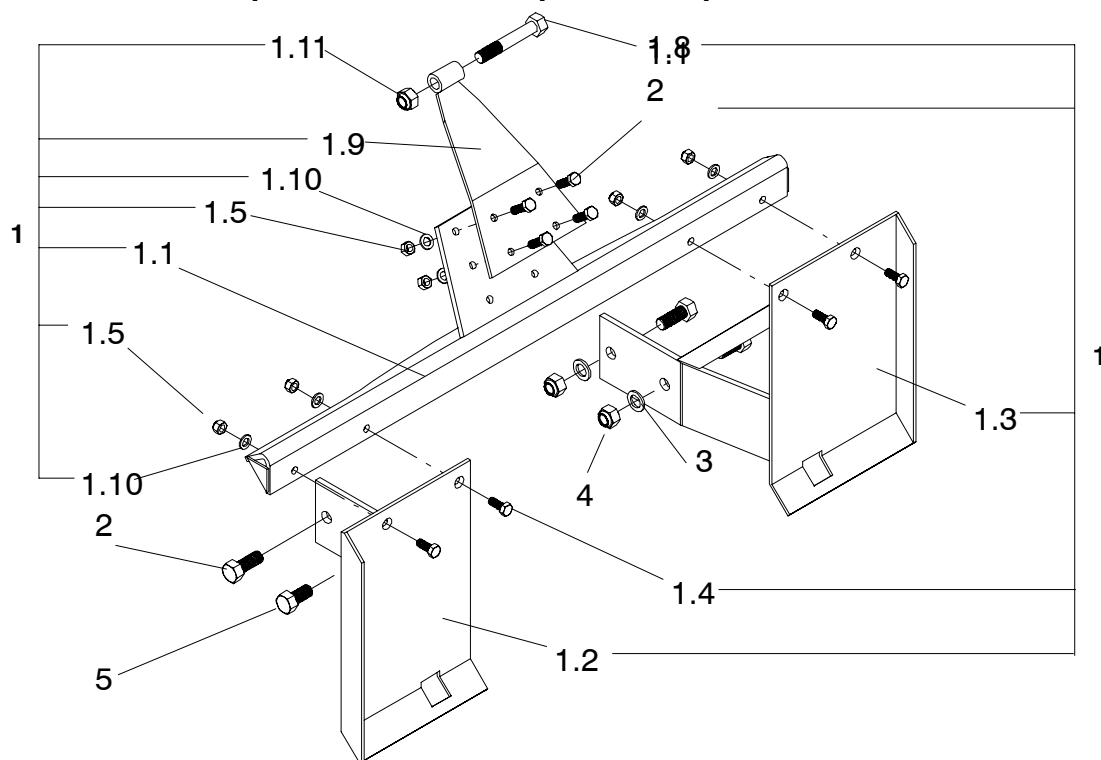


43510500

Part Osa Teil Pièce	Order No. Til.nro. Best Nt No. de ordre	Description Bezeichnung	Nimitys Désignation	Remarks Huom. Obs Obs	Qty. Kpl. St. Pc.
	03510500	Adapter Wille,complete Adaption Wille, Komplet	Sovitus wille,täydellinen Adapteur Wille,complete		
1.	43510490	Frame Rahmen	Runko Châssis		1
2.	43510510	Plate Platte	Levy Accrocheur		2
3.	52062163	Hexagonal screw Sechskanntschraube	Kuusioruuvi Vis hexagonale	M16x100 DIN931	4
4.	52117165	Lock nut Schlossmutter	Lukitusmutteri Ecrou de verrouillage	M16 DIN985	4
5.	52200078	Washer Scheibe	Aluslevy Rondelle	M16 DIN126	4
6.	43512510	Hexagonal screw Sechskanntschraube	Kuusioruuvi Vis hexagonale	M20x120	1
7.	52117207	Lock nut Schlossmutter	Lukitusmutteri Ecrou de verrouillage	M20 DIN985	1

CH 160

Adapter – Sovitus – Adaption – Adaptateur Bobcat

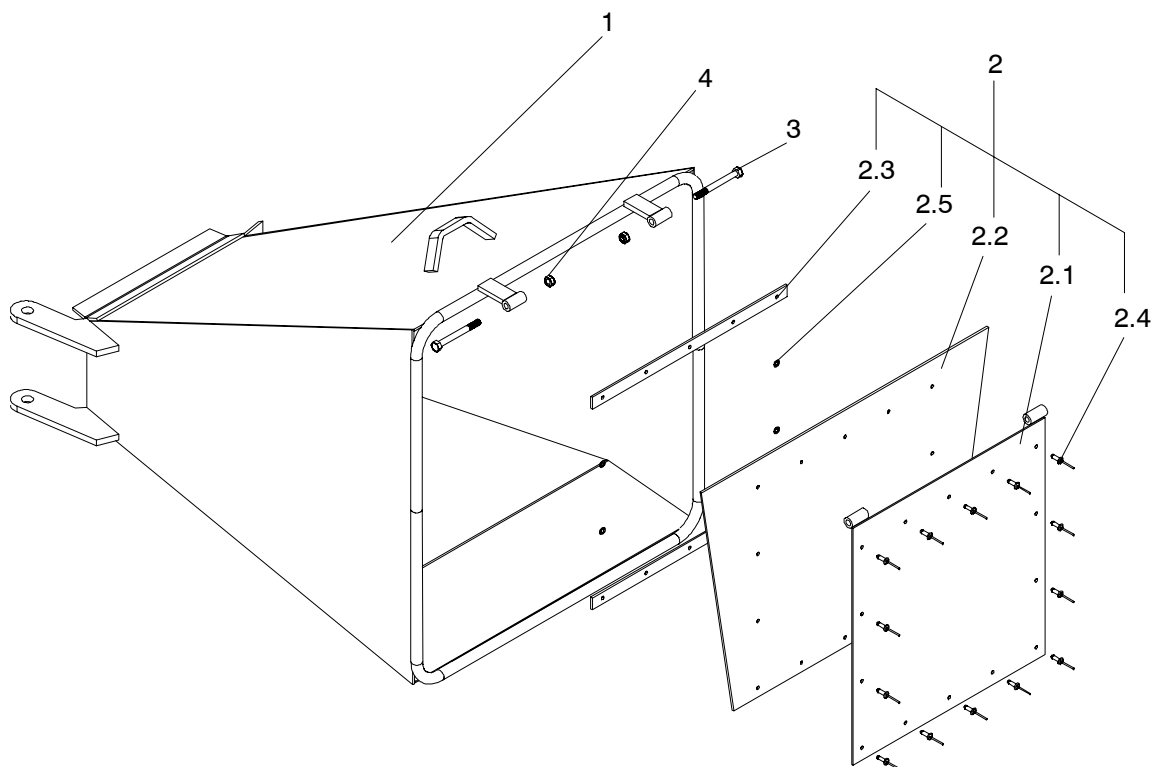


03510800

Part Osa Teil Pièce	Order No. Til.nro. Best Nt No. de ordre	Description Bezeichnung	Nimitys Désignation	Remarks Huom. Obs Obs	Qty. Kpl. St. Pc.
1.	03510800 ...	Adapter frame Adaption Rahmen	Sovitusrunko Adaptateur Bobcat		1
1.1. ...	43510820 ...	Upper beam Ober Balken	Yläpalkki Poutre supérieure		1
1.2. ...	43510830 ...	Fastener, left Verschluss, Link	Kannatin, vasen Support, gauche		1
1.3. ...	43510840 ...	Fastener, right Verschluss, Rechte	Kannatin, oikea Support, droit		1
1.4. ...	52062023 ...	Hexagonal screw Sechskanntschaube	Kuusioruuvi Boulon hexagona	M12x30 DIN933	4
1.5. ...	52117124 ...	Lock nut Schlossmutter	Lukitusmutteri Ecroû de verrouillage	M12 DIN985	8
1.6. ...	–				
1.7. ...	–				
1.8. ...	43512510 ...	Hexagonal screw Sechskanntschaube	Kuusioruuvi Boulon hexagona	M20x120	1
1.9. ...	43513120 ...	Fastener plate Verschlussplatte	Kiinnityslevy Plaque de montage		1
1.10.	52200466 ...	Washer Scheibe	Aluslevy Rondelle	M12 DIN126	8
1.11.	52117207 ...	Lock nut Schlossmutter	Lukitusmutteri Ecroû de verrouillage	M20 DIN985	1
1.12.	52062031 ...	Hexagonal screw Sechskanntschaube	Kuusioruuvi Vis hexagonale	M12x40 DIN933	4
2.	52062221 ...	Hexagonal screw Sechskanntschaube	Kuusioruuvi Vis hexagonale	M20x50 DIN933	2
3.	52211083 ...	Spring washer Federscheibe	Jousialuslevy Rondelle frein	M20 DIN127	4
4.	52117207 ...	Lock nut Schlossmutter	Lukitusmutteri Ecroû de verrouillage	M20 DIN985	4
5.	52062213 ...	Hexagonal screw Sechskanntschaube	Kuusioruuvi Vis hexagonale	M20x40 DIN933	2

CH 160

F150

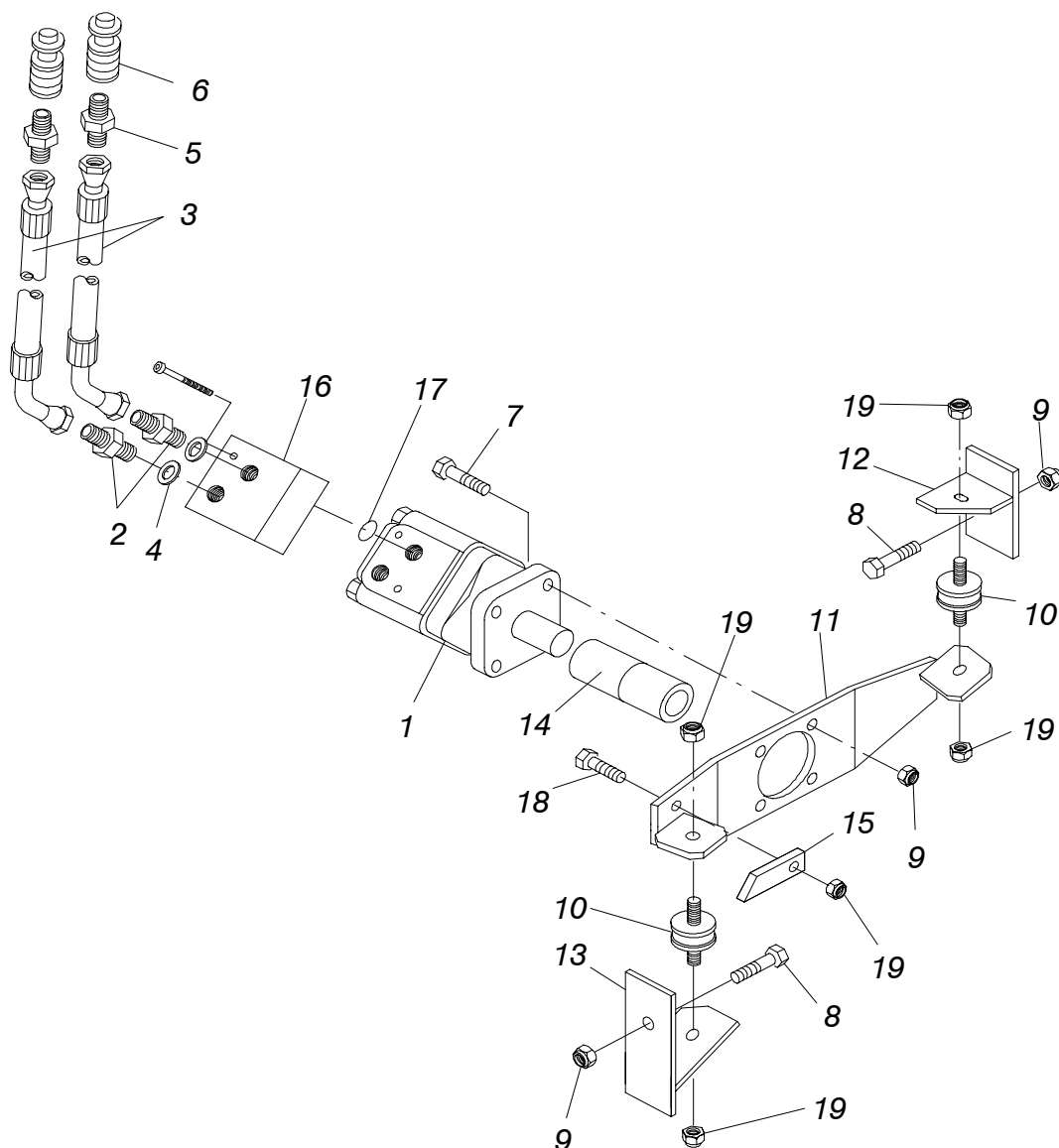


03342144

Part Osa Teil Pièce	Order No. Til.nro. Best Nt No. de ordre	Description Bezeichnung	Nimitys Désignation	Remarks Huom. Obs Obs	Qty. Kpl. St. Pc.
	03342144	F150 Mechanical feed chute F150 Verlängerungsteile	F150 mekaaninen syöttölaite F150 Goulotte d'alimentation mécanique		
1.	03342151	Feed chute Zuführtrichter	Syöttösuppilo Goulotte d'alimentation		1
2.	43340876	Flap, complete Klappe, Komplet	Läppä, täydellinen Glissière, complète		1
2.1.	43340884	Flap Klappe	Läppä Glissière		1
2.2.	43340900	Flap rubber Gummiklappe	Kumiläppä Tampon en caoutchouc de glissière	570x600	1
2.3.	43340918	Support Platte	Latta Petite courroie d'extrémité		2
2.4.	52832094	Rivet Niete	Niitti Rivet explosif	4,8x15 Fe	14
2.5.	52200011	Washer Scheibe	Aluslaatta Rondelle	M5 DIN126	4
3.	03342151	Hexagonal screw Sechskanntschraube	Kuusioruuvi Vis hexagonale	M8x80 DIN931	2
4.	52117085	Lock nut Schlossmutter	Lukitusmutteri Ecrou de verrouillage	M8 DIN985	2

CH 160

HD 100

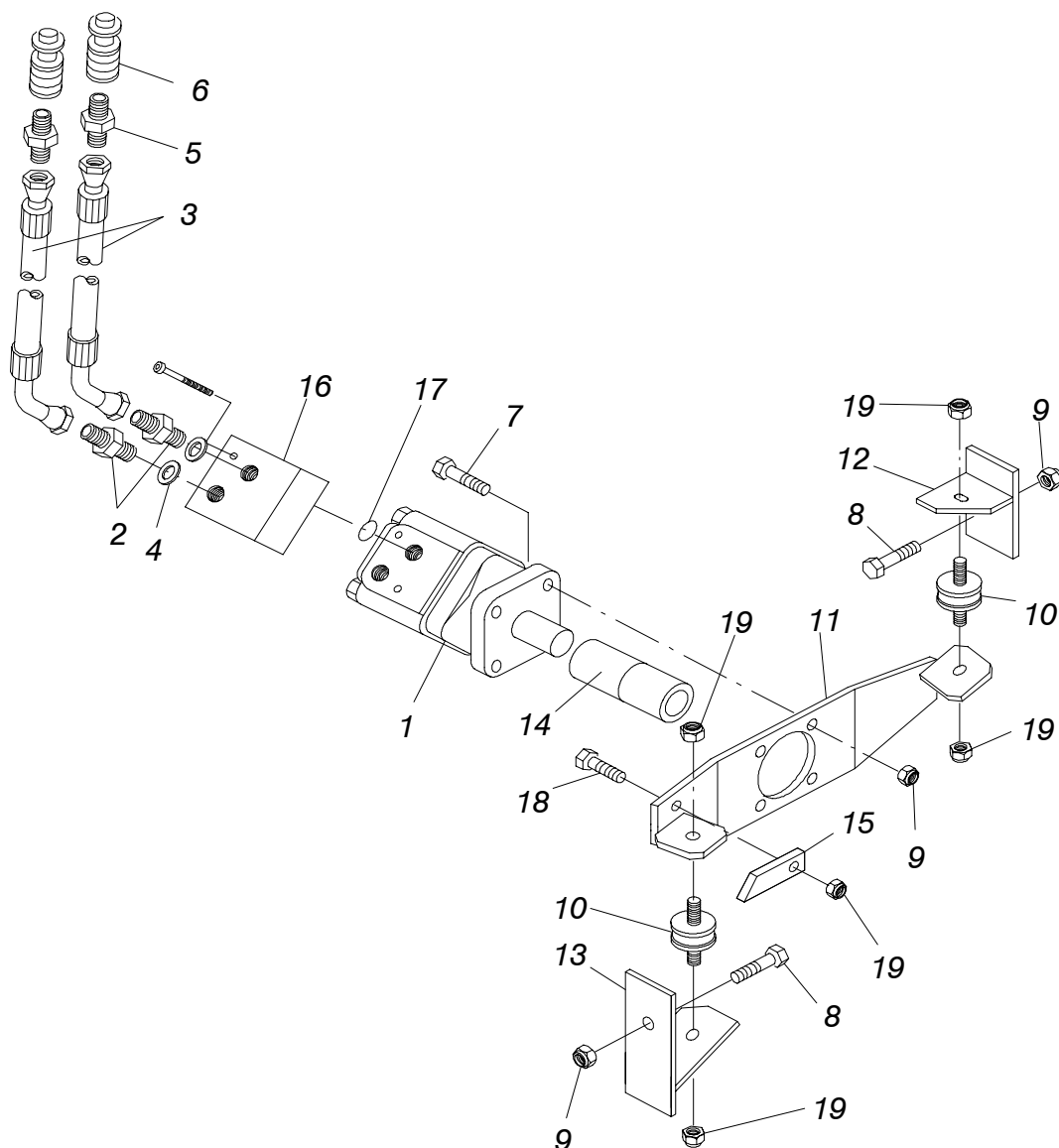


03510700

Part Osa Teil Pièce	Order No. Til.nro. Best Nt No. de ordre	Description Bezeichnung	Nimitys Désignation	Remarks Huom. Obs Obs	Qty. Kpl. St. Pc.
1.	56001589 ...	Hydraulic motor Hydraulikmotor	Hydraulimoottori Moteur hydraulique	Lösi EPMS 100 C	1
2.	52432044 ...	Double fitting Doppelnippel	Kaksoisnippa Double embout	R1/2"-3/4"	2
3.	56533151 ...	Hose assy Schlauchsatz	Letkuasennelma Ensemble de tuyaux	V3/4"S L=1,0m	2
4.	52390580 ...	Usit-ring Usit-Dichtung	Usit-rengas Bague Usit	U27,2x36,0x2,0	2
5.	52449220 ...	Quick fitting Schnappkupplung	Pikaliitinnippa Embout douille à baïonnette	3/4 (Sk)	2
6.	52449238 ...	Quick fitting coupling Schnappkupplung	Pikaliitinrunko Corps de la douille baïonnette		2
7.	52062031 ...	Hexagonal screw Sechskanntschaube	Kuusioruuvi Vis hexagonale	M12x40 DIN933 8.8Zn	4
8.	52062023 ...	Hexagonal screw Sechskanntschaube	Kuusioruuvi Vis hexagonale	M12x30 DIN933 8.8Zn	2
9.	52117124 ...	Lock nut Schlossmutter	Lukistusmutteri Ecrou de verrouillage	M12 DIN985 8.8Zn	14
10.	54923073 ...	Damper Dämpfer	Tasovaimennin Amortisseur de vibrations	17/1388 1MN60	2

CH 160

HD 100

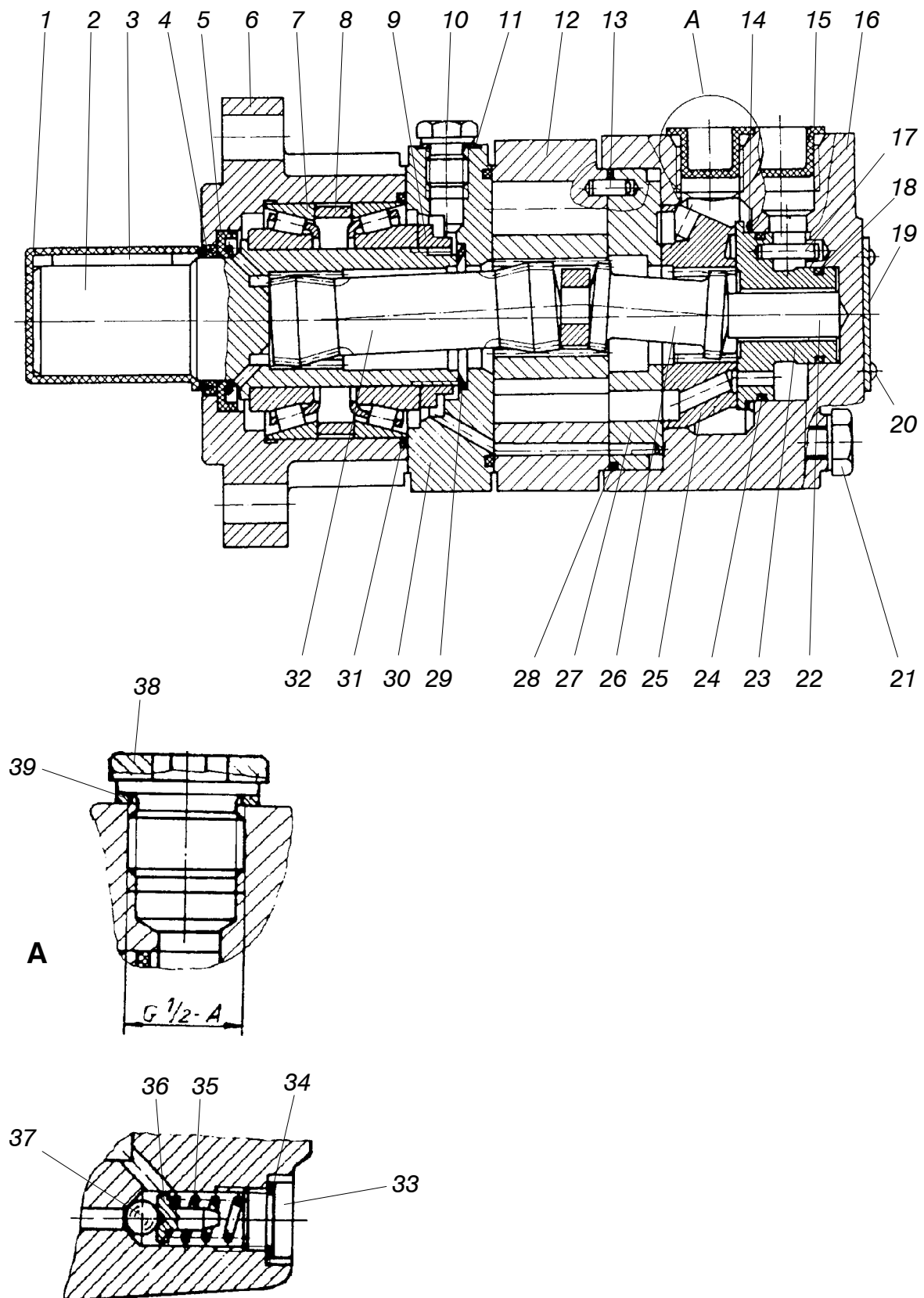


03510700

Part Osa Teil Pièce	Order No. Til.nro. Best Nt No. de ordre	Description Bezeichnung	Nimitys Désignation	Remarks Huom. Obs Obs	Qty. Kpl. St. Pc.
11.	43510640 ...	Motor support Motorträger	Moottorin kannake Support-moteur		1
12.	43510620 ...	Support, left Träger, Links	Kannake, vasen Support, gauche		1
13.	43510630 ...	Support, right Träger, Rechts	Kannake, oikea Support, droit		1
14.	43510590 ...	Coupling sleeve Kupplungshülse	Kytkinholkki Manchon d'accouplement		1
15.	43510540 ...	Limitter Platte	Rajoitinlevy Plaque de butée	10x25	1
16.	56053986 ...	Valve Valve	Venttiili Soupape d'admission		1
17.	52301397 ...	O-ring O-Ring	O-rengas Joint torique	21,95x1,78 SH	2
18.	52060225 ...	Hexagonal screw Sechskanntschaube	Kuusioruuvi Vis hexagonale	M10x25 DIN933 8.8Zn	1
19.	52117108 ...	Lock nut Schlossmutter	Lukitusmutteri Ecroû de verrouillage	M10 DIN985	5

CH 160

LÖSI 100



56001589

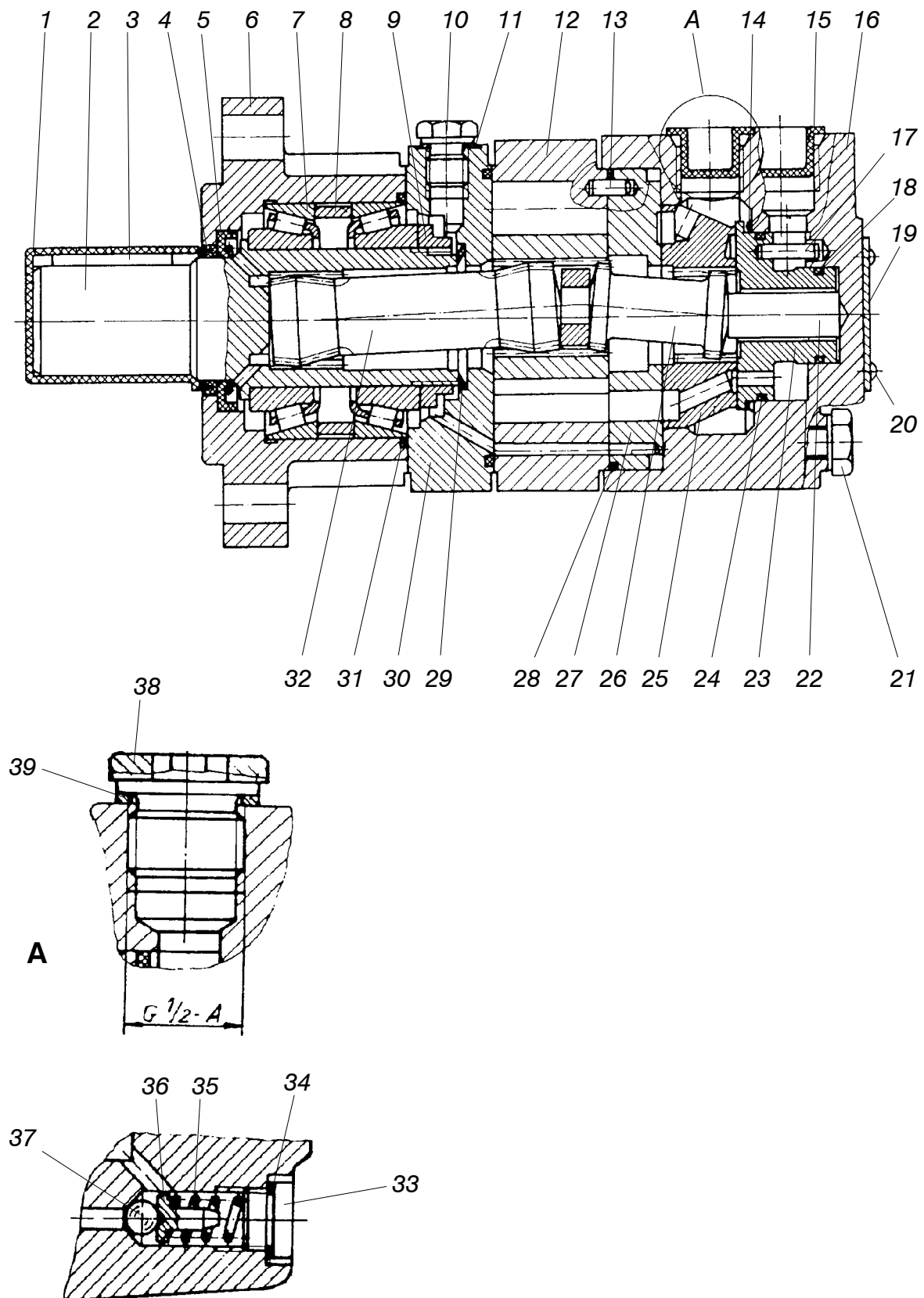
CH 160

LÖSI 100

Part Osa Teil Pièce	Order No. Til.nro. Best Nt No. de ordre	Description Bezeichnung	Nimitys Désignation	Remarks Huom. Obs Obs	Qty. Kpl. St. Pc.
	56001589 ...	Hydraulic motor Hydraulik motor	Hydraulimootori Moteur hydraulique	Lösi EPMS 100 C	
1.		Axle cover Schutzdeckel	Akselin suoja Bouchon de protection pour l'arbre		1
2.	58005307 ...	Axle Achse	Akseli Axe		1
3.	58005315 ...	Key Keil	Kiila Coin	A10x8x45 DIN6885	1
4.		Seal Wellendichtung	Tiivisterengas Bague d'étanchéité	35x42x4	1
5.		Seal Wellendichtung	Akselitiiviste Garniture d'étanchéité d'arbre	35x52x5,5	1
6.		Housing Gehäuse	Kotelo Logement		1
7.		Bearing Lager	Laakeri Palier		2
8.		Bearing Lager	Laakeri Palier		1
9.		Lock nut Schlossmutter	Lukitusmutteri Ecrou de verrouillage	M40x1,5	1
10.		Plug Propfen	Tulppa Fiche	G1/4"–A	1
11.		Washer Scheibe	Aluslevy Rondelle	13x17	1
12.		Sprocket kit Zahnradatz	Hammaspyöräsarja Ensemble roue d'engrenage		1
13.		Pin Zapfen	Tappi Cheville	3.175x12.7	1
14.		Spring washer Federscheibe	Jousialuslevy Rondelle frein		1
15.		Seal plug Dichtungspropfen	Tiivistetulppa Bouchon d'étanchéité		2
16.		End plate Endflansch	Päätylaippa Bride pleine		1
17.		Pin Zapfen	Tappi Cheville	4.72x20	1
18.		O–ring O–Ring	O–rengas Joint torique	24x2	1
19.		Label Namensschild	Nimilaatta Plaque signalétique		1
20.		Rivet Niete	Niitti Clou à river	2x5	4

CH 160

LÖSI 100

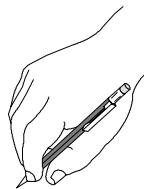


56001589

CH 160

LÖSI 100

Part Osa Teil Pièce	Order No. Til.nro. Best Nt No. de ordre	Description Bezeichnung	Nimitys Désignation	Remarks Huom. Obs Obs	Qty. Kpl. St. Pc.
21.		Screw Schraube	Ruuvi Boulon	M10x130	4
22.		Plate Distanzplatte	Välilevy Entretoise	L=30,5mm	1
23.		Plate Stabilisierungsplatte	Tasapainoituslevy Plaque compensateur		1
24.		O-ring O-Ring	O-rengas Joint torique	45x2	1
25.		Valve plate Ventilplatte	Venttiililevy Plaque porte-soupape		1
26.		Valve guide Ventilbenutzer	Venttiilin ohjain Entraîneur de soupape		1
27.		Plate Platte	Levy Plaque		1
28.		O-ring O-Ring	O-rengas Joint torique	82x2	2
29.		Seal ring Wellendichtung	Tiivisterengas Bague d'étanchéité		1
30.		Plate Platte	Levy Plaque		1
31.		O-ring O-Ring	O-rengas Joint torique	69x3	1
32.		Coupling sleeve Kardanwelle	Kardaaniakseli Arbre de relais	L=73mm	1
33.		Plug Propfen	Tulppa Fiche	M10x1	2
34.		Washer Scheibe	Aluslevy Rondelle	A10x14	2
35.	58005646	Guide pin Steuerzapfen	Ohjaintappi Doigt de guidage		2
36.	58005646	Guide pin Steuerzapfen	Ohjaintappi Doigt de guidage		2
37.	58005653	Ball Kugel	Pallo Balle	6.35	2
38.		Plug Propfen	Tulppa Fiche	G-1/2"-A	2
39.		Washer Scheibe	Aluslevy Rondelle	22x27 - Cu	2
40.	58005869	Seal kit Dichtungssatz	Tiivistesarja Nécessaire d'étanchéité	4,5,18,24,28,29,31	1



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GUARANTEE

FARMI Forest products are guaranteed to be free from defects in materials and workmanship for 12 months from the date of purchase.

This warranty does not cover defects resulting from:

- misuse of the product
- inadequate maintenance
- modifications of the product made without the manufacturer's consent

The warranty is effective only, if the lower portion of this page is completed and returned to the manufacturer within 14 days of receipt of the product..

Important

Before putting your FARMI to use, be sure to read the instruction and service booklet provided with the machine

FARMI Forest Corporation

Ahmolantie 6

74510 PELTOSALMI



FARMI Forest

Ahmolantie 6

FIN – 74 510 PELTOSALMI

FINLAND



Date of delivery: ____ / ____ 20 ____

Dealer: _____

Dealer's address: _____

Dealer's tel: _____

Product and type: _____

Serial number: _____



Return to the manufacturer

Date of delivery: ____ / ____ 20 ____

Dealer: _____

Dealer's address: _____

Dealer's tel: _____

Customer: _____

Customer's address: _____

Customer's tel: _____

Product and type: _____

Serial number: _____

Notes: _____



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Forest
Made in Finland

